PHYSICS

1) The pressure P and volume V of a gas are related as $PV^{3/2} = K$, where K is a constant. The percentage increase in the pressure for a reduction of 0.5% of the volume will be :

(1) 0.25%
(2) 0.75%
(3) 1.50%
(4) 0.50%

2) Velocity-time equation of a particle moving in a straight line is v = 2t – 4 for $t \leqslant 2s$ and v = 6 – 3t for $t \geqslant 2s$, where v is in m/s and t in seconds. The distance travelled by the particle in the time interval from t = 0 to t = 4s is:

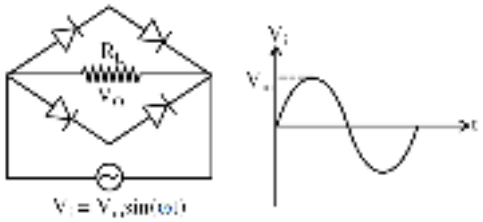
(1) 12 m
(2) 16m
(3) 4m
(4) 10 m

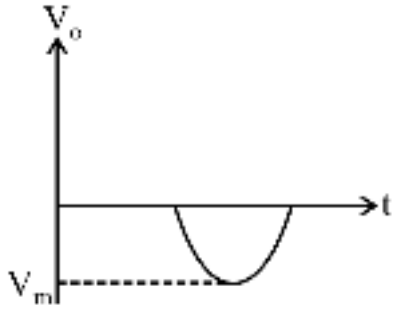
3) A stone is thrown up at an angle α to the horizontal from a tower of height h. It hits the ground at distance d from the foot of the tower. Find the time of flight :

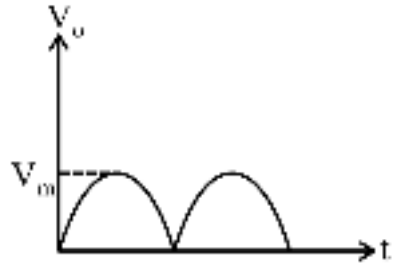
(1) $\sqrt{\frac{2(h + d\tan\alpha)}{g}}$
(2) $\sqrt{\frac{2d\tan\alpha}{g}}$
(3) $\sqrt{\frac{2h}{g}}$
(4) $\sqrt{\frac{2(h + d)}{g}}$

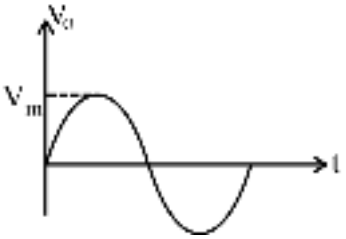
0

4) In the given circuit, output wave form v will be :-

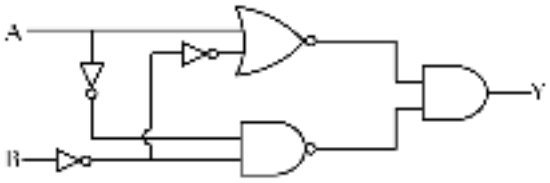


(1) 

(2) 

(3) 

(4) None of these

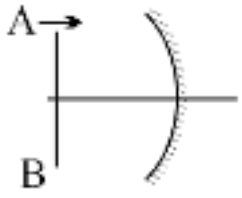
5) For given combination of logic gates, output Y will be :- 

(1) 0

(2) 1

(3) $\overline{A.B}$

(4) A + B

6) A plane wavefront AB is incident on a concave mirror as shown. Then, the wavefront just after reflection is : 

(1) 

(2) 

(3) 

(4) None of the above

7) In a double slit experiment, when a thin film of thickness t having refractive index μ is introduced in front of one of the slits, the maximum at the centre of the fringe pattern shifts by one fringe width. The value of t is (λ is the wavelength of the light used) :

(1) $\frac{\lambda}{2(\mu-1)}$

(2) $\frac{\lambda}{(2\mu-1)}$

(3) $\frac{2\lambda}{(\mu-1)}$

(4) $\frac{\lambda}{(\mu-1)}$

8) **Assertion (A) :** Stopping potential is minimum positive potential of Anode with respect to cathode at which photo current becomes zero.
Reason (R) : At stopping potential fastest electron is stopped to reach collector.

(1) Both (A) & (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) & (R) are correct but (R) is not correct explanation of (A).

(3) Both (A) & (R) are incorrect.

(4) (A) is incorrect while (R) is correct.

9) Read following statements carefully and select correct statements.
(a) Kinetic energies of emitted photo electrons are not equal.
(b) Electrons are in different energy levels in a metal.
(c) Work function is related with threshold frequency.
(d) Time interval between incident of photon and emission of electron is 10^{-9} s.

(1) Only a, b

(2) a, b, c

(3) b, c, d

(4) All of the above

10) Two balls of mass 2kg and 4kg are dropped from top of a tower simultaneously. Find ratio of their de-Broglie wavelength when they have fallen trough 80 m :-

(1) 1 : 2

(2) 4 : 1

(3) 2 : 1

(4) 1 : 4

11) Electrons of wavelength 0.01Å incident on a metal, then cut off wavelength of emitted x-rays will

be :

(1) 8.26×10^{-3} Å

(2) 5.3×10^{-2} Å

(3) 1.23×10^{-6} Å

(4) 7.32×10^{-4} Å

12) In a compound microscope, the focal length of objective and eyepiece are 1.5 cm and 6.25 cm respectively. If an object is placed at 2 cm from objective and the final image is formed at 25 cm from eye lens, the distance between two lenses is -

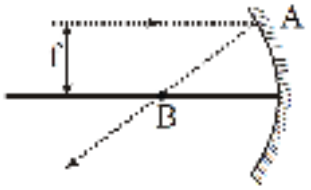
(1) 6.00 cm

(2) 7.75 cm

(3) 9.25 cm

(4) 11.0 cm

13) A ray of light is incident on a concave mirror. It is parallel to the principal axis and its height from principal axis is equal to the focal length of the mirror. The ratio of the distance of point B from the centre of curvature, to the focal length is (AB is the reflected ray) :-



(1) $\frac{2}{\sqrt{3}}$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{2}{3}$

(4) $\frac{1}{2}$

14) A point source of light is placed 4 m below the surface of water of $\mu = \frac{5}{3}$. The minimum diameter of a disc (in m), which should be placed over the source, on the surface of water to cut off all light coming out of water is:

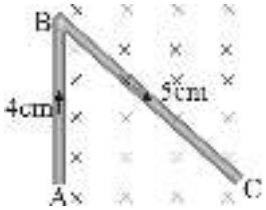
(1) 6

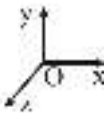
(2) 3

(3) 2

(4) 3.5

15) A uniform conducting wire ABC has a mass of 10 g. A current of 2A flows through it. The wire is kept in a uniform magnetic field B = 2T. The acceleration of the wire will be :-





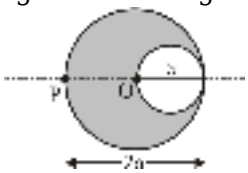
(1) Zero

(2) 12 ms^{-2} along y-axis

(3) 1.2×10^{-3} ms^{-2} along y-axis

(4) 0.6×10^{-3} ms^{-2} along y-axis

16) A cylindrical cavity of diameter a exists inside a cylinder of diameter 2a as shown in the figure. Both the cylinder and the cavity are infinitely long. A uniform current density J flows along the length. If the magnitude of the magnetic field at the point P is given by $\frac{N}{12}\mu_0 aJ$, then the value of N is



(1) 2

(2) 3

(3) 4

(4) 5

17) An ideal choke takes a current of 10 amp when connected to an AC supply of 125 volt and 50 Hz. A pure resistor under the same conditions takes a current of 12.5 amp. If the two are connected to an AC supply of 100 $\sqrt{2}$ volt and 40 Hz, then the current in series combination of above resistor and inductor is :-

(1) 10 amp

(2) 12.5 amp

(3) 20 amp

(4) 25 amp

18) **Assertion :** Velocity of light-depends on electric and magnetic properties of the medium.

Reason : Velocity of light in a medium is given by $v = \frac{1}{\sqrt{\mu \in}}$

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

19) Two particles perform simple harmonic motion of same frequency and about same mean position. Their amplitude is same and is equal to A and their time period is T. If at t = 0 their

separation is maximum and equal to A, their separation at t = $\frac{T}{12}$ is

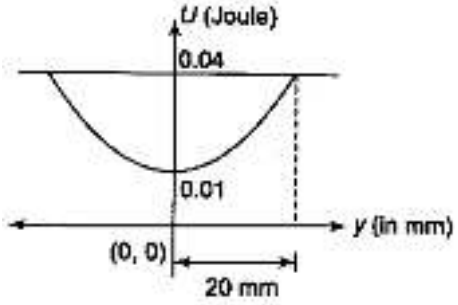
(1) $\frac{\sqrt{3}}{2}A$

(2) A

(3) $\frac{A}{\sqrt{2}}$

(4) $\frac{A}{2}$

20) The variation of potential energy of a harmonic oscillator is as shown in figure. The spring constant is



(1) 1×10^2 N/m

(2) 150 N/m

(3) 0.667×10^2 N/m

(4) 3×10^2 N/m

21) Ratio of slope of P-V curve of adiabatic process for diatomic gas and monoatomic gas at pressure P and volume V is :

(1) 1

(2) $\frac{5}{7}$

(3) $\frac{7}{15}$

(4) $\frac{21}{25}$

22) One mole of a diatomic gas undergoes a process $P = \frac{P_0}{\left(1 + \frac{V}{V_0}\right)^3}$, where P_0, V_0 are constants. The translational kinetic energy of the gas when $V = V_0$, is given by :

(1) $\frac{3P_0V_0}{2}$

(2) $\frac{3P_0V_0}{16}$

(3) $\frac{3P_0V_0}{4}$

(4) $\frac{5P_0V_0}{2}$

23) There is formation of layer of ice y cm thick on water, when the temperature of air is –θ°C (less than freezing point). The thickness of layer increases from y to x in the time t, then the value of t is given by

(1) $\frac{(x+y)(x-y)\rho L}{2k\theta}$

(2) $\frac{(x-y)\rho L}{2k\theta}a$

(3) $\frac{(x+y)(x-y)\rho L}{k\theta}$

(4) $\frac{(x-y)\rho Lk}{2\theta}$

24) At what temperature is the rms velocity of a hydrogen molecule equal to that of an oxygen molecules at 367ºC?

(1) 80 K

(2) 40 K

(3) 60 K

(4) 100 K

25) An open pipe is in resonance in its 2nd harmonic with tuning fork of frequency f_1. Now it is closed at one end and the frequency of the tuning fork is increased slowly from f_1, then again a resonance is obtained with a frequency f_2. If in this case the pipe vibrates n^{th} harmonic, then select the correct option :

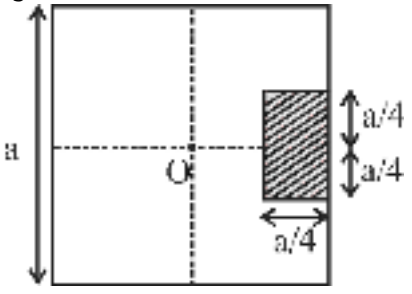
(1) $n = 3, f_2 = \frac{3}{4} f_1$

(2) $n = 3, f_2 = \frac{5}{4} f_1$

(3) $n = 5, f_2 = \frac{5}{4} f_1$

(4) $n = 5, f_2 = \frac{3}{4} f_1$

26) A uniform square sheet is given. A part of it of given dimension is cut from it, then distance of centre of mass of remaining part from O will be :-



(1) $\frac{a}{8}$

(2) $\frac{3a}{8}$

(3) $\frac{3a}{56}$

(4) $\frac{3a}{64}$

27) A smooth sphere is moving on a horizontal smooth surface with velocity vector $2\hat{i}+2\hat{j}$ immediately before it hits a vertical wall. The wall is parallel to $\hat{j}$ vector and the coefficient of restitution between the sphere and the wall is $e=\frac{1}{2}$. The velocity vector of the sphere after it hits the wall is:-

(1) $\hat{i}-\hat{j}$

(2) $-\hat{i}+2\hat{j}$

(3) $-\hat{i}-\hat{j}$

(4) $2\hat{i}-\hat{j}$

28) A spherical soap bubble of radius 3 cm is formed inside another spherical soap bubble of radius 6 cm. If the internal pressure of the smaller bubble of radius 3 cm in the above system is equal to the internal pressure of the another single soap bubble of radius r cm. The value of r is.....

(1) 2 cm

(2) 3 cm

(3) 4 cm

(4) 5 cm

29) A brass rod of cross-sectional area 1 cm^2 and length 0.2 m is compressed lengthwise by a weight of 5 kg. If Young's modulus of elasticity of brass is 1×10^{11} N/m^2 and g = 10 m/sec^2, then increase in the potential energy of the rod will be :-

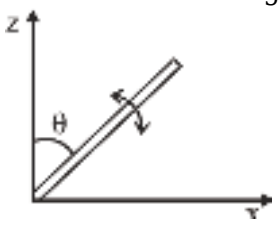
(1) 10^{-5} J

(2) 2.5×10^{-5} J

(3) 5×10^{-5} J

(4) 2.5×10^{-4} J

30) A uniform rod of mass M and length is pivoted at one end so that it can rotate in a vertical plane (see figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. The angular acceleration of the rod, when it makes an angle θ with the vertical, is :



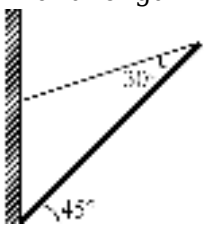
(1) $\frac{3g}{2\ell}\cos\theta$

(2) $\frac{2g}{3\ell}\cos\theta$

(3) $\frac{3g}{2\ell}\sin\theta$

(4) $\frac{2g}{3\ell}\sin\theta$

31) A rod of mass M and length L with the help of rope has held as shown in figure, then tension in the rope will be :-



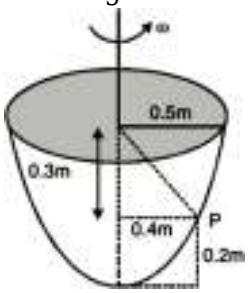
(1) $\frac{Mg}{\sqrt{2}}$

(2) $\frac{Mg}{3}$

(3) $\frac{Mg}{2}$

(4) $\frac{2Mg}{3}$

32) As shown in figure, a particle P will be in equilibrium inside a hemispherical bowl of radius 0.5 m at a height 0.2 m from the bottom when the bowl is rotated at an angular speed ($g = 10 m/sec^2$):



(1) $10/\sqrt{3}$ rad/sec

(2) $10\sqrt{3}$ rad/sec

(3) 10 rad/sec

(4) $\sqrt{20}$ rad/sec

33) Two uniform solid spheres of equal radii R, but mass M and 4M have a centre to centre separation 9R. Both the spheres are held fixed. A projectile of mass m is projected from the surface of the sphere of mass M directly towards the centre of the second sphere. Then, find the minimum kinetic energy of projectile required to throw it so that it can reach the surface of second sphere :-

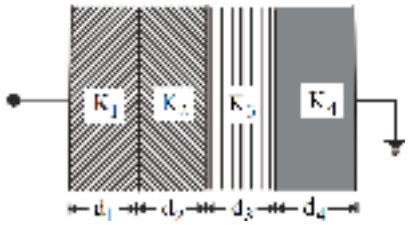
(1) $\frac{GMm}{R}$

(2) $\frac{3}{2}\frac{GMm}{R}$

(3) $\frac{5}{2}\frac{GMm}{R}$

(4) $\frac{GMm}{2R}$

34) The effective capacity of the system in adjoining figure will be :-



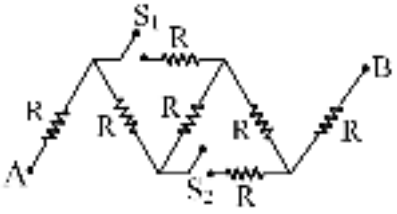
(1) $C = \frac{\epsilon_0 A}{\left[\frac{d_1}{K_1} + \frac{d_2}{K_2} + \frac{d_3}{K_3} + \frac{d_4}{K_4}\right]}$

(2) $C = \frac{\epsilon_0 A}{4d}$

(3) $C = \frac{4d}{\epsilon_0 A}$

(4) $C = \frac{K_1 K_2 K_4 K_3}{4d}$

35) Find the ratio of resistance between A & B before and after switch (S_1 & S_2) are closed in the given circuit :



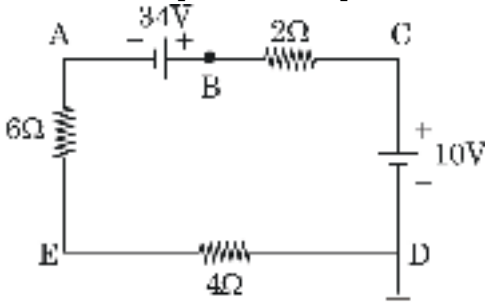
(1) 5 : 4

(2) 4 : 5

(3) 5 : 3

(4) 5 : 6

36) Find the potential at point E, if D is grounded, as shown in figure.



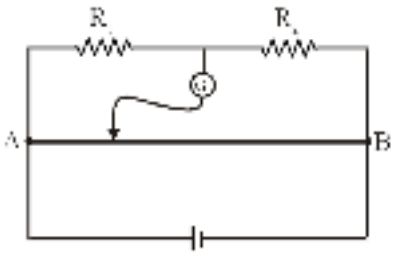
(1) 12 volt

(2) –8 volt

(3) 14 volt

(4) –20 volt

37) In the figure shown, for given values of R_1 and R_2, the balance point for Jockey is at 40 cm from A. When R_2 is shunted by a resistance of 10Ω, balance point shifts to 50 cm. R_1 and R_2 are (AB = 1m):



(1) $\frac{10}{3}\Omega$, 5Ω

(2) 20Ω, 30Ω

(3) 10Ω, 15Ω

(4) 5Ω, $\frac{15}{2}\Omega$

38) Dimensional formula of [β] in $\frac{\alpha}{t^2} = FV + \frac{\beta}{x^2}$ is
(T = Time, F = Force, V = Velocity, X = Distance)

(1) $[M^1L^4T^{-3}]$

(2) $[M^2L^4T^{-2}]$

(3) $[M^2L^3T^{-3}]$

(4) $[M^2L^3T^{-2}]$

39) The pitch of screwgauge is 1 mm and number of circular scale divisions in screwgauge are 50. The screwgauge reads 3 circular divisions when nothing is put in-between jaws of screwgauge. By using this screwgauge, the diameter of wire is measured, the reading is 7 divisions on linear scale and 36 division on circular scale. The diameter of wire is :

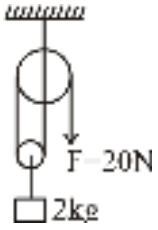
(1) 7.66 mm

(2) 7.78 mm

(3) 7.72 mm

(4) 7.33 mm

40) The acceleration of the 2 kg block, if the free end of string is pulled with a force of 20 N as shown is -



(1) Zero

(2) 10 m/sec^2

(3) 5 m/sec^2 upward

(4) 5 m/sec^2 downward

41) 80 railway wagons all of same mass 5×10^3 kg are pulled by engine with a force of 4×10^5N. The tension in the coupling between 30th and 31st wagon from the engine is -

(1) 25×10^4 N

(2) 40×10^4 N

(3) 20×10^4 N

(4) 32×10^4 N

42) A block of mass 10 kg moving in x-direction with constant speed of 10 m/s is subjected to the retarding force F = 0.2 x during its travel from x = 0 m to x = 10 m, then its final KE will be :

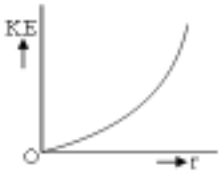
(1) 490 J

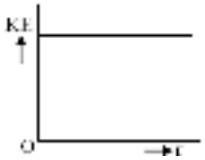
(2) 275 J

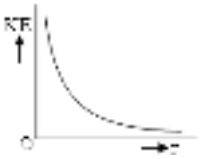
(3) 510 J

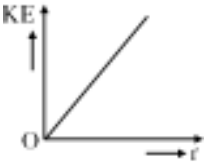
(4) 475 J

43) An infinitely long positively charged straight thread has a linear charge density λ. An electron revolves along a circular path having axis along the length of the wire. The graph that correctly represents the variation of the kinetic energy of electron as a function of radius of circular path from the wire is :

(1) 

(2) 

(3) 

(4) 

44) Two fixed charges A and B of 5 μC each are separated by a distance of 6m. C is the mid point of the line joining A and B. A charge 'Q' of –5μC is shot perpendicular to the line joining A and B from C with a kinetic energy of 0.06 J. The charge 'Q' comes to rest at a point D. The distance CD is :-

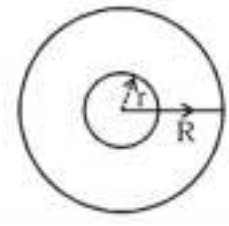
(1) 3 m

(2) $\sqrt{3}$m

(3) $3\sqrt{3}$m

(4) 4 m

45) Which of the following statement is/ are true for two concentric coplanar coil shown:



Statement 1 : we can find mutually induced EMF and mutual inductance when current only in small coil is varying with time **Statement 2 :** reciprocity theorem states that $M_{12} = M_{21}$, that is mutual inductance is purely geometrical quantity depending only on size, number of turns, relative position and configuration of two coils but independent of current.

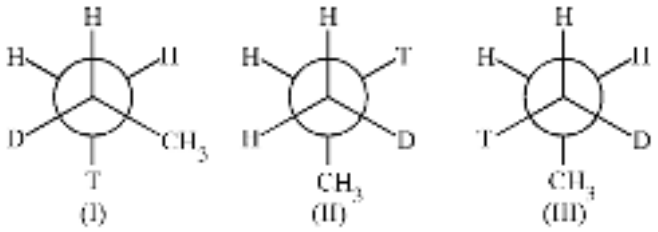
(1) Only statement 1 is correct.

(2) Only statement 2 is correct.

(3) Both statement 1 and statement 2 are correct.

(4) None

CHEMISTRY

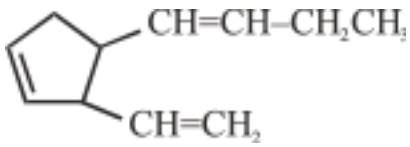
1) 

(1) I and II are identical

(2) II and III are chain isomers

(3) I and III are positional isomers

(4) II and III are identical

2) Total number of stereo isomer of given compound 

(1) 8

(2) 32

(3) 4

(4) 16

3) Match the column

	Column-I (Mixture)		**Column-II (Purification technique)**
A	Chloroform & aniline	P	Distillation
B	Aniline & water	Q	Chromatography

C	Amino acids	R	Steam distillation
D	Glycerol	S	Distillation under reduced pressure

Which of the following is correct.

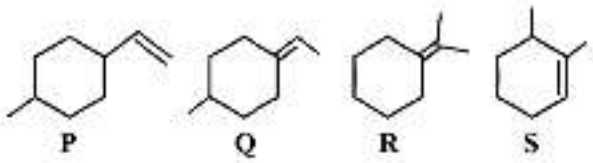
(1) A-P; B-Q; C-R; D-S

(2) A-P; B-S; C-R; D-Q

(3) A-P; B-R; C-Q; D-S

(4) A-R; B-S; C-P; D-Q

4) Arrange the following alkenes in increasing order of their heat of hydrogenation :-

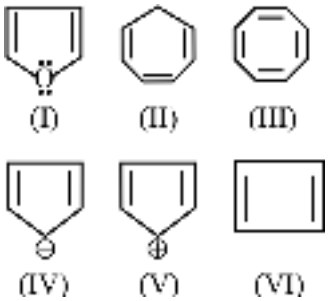


(1) R < Q < S < P

(2) R < S < P < Q

(3) P < Q < R < S

(4) P < Q < S < R



5) Which of the following compounds are anti-aromatic ?

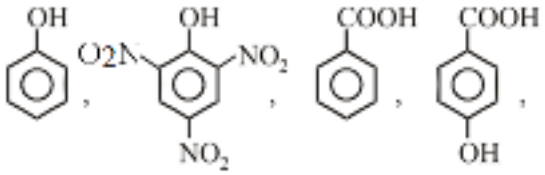
(1) (III) and (VI)

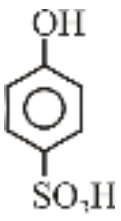
(2) (II) and (V)

(3) (I) and (V)

(4) (V) and (VI)

6) How many of the following will give $NaHCO_3$ test? [CO_2 effervesence will be observed]





HCOOH, CH_3CH_2OH, CH_3–C≡CH,

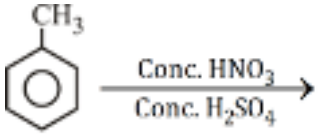
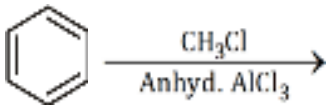
(1) 4

(2) 5

(3) 6

(4) 3

7) Match the following :-

	Reaction Column-I		**Name of Reaction Column-II**
(a)		(p)	Kolbe's electrolysis
(b)	CH_3–COOK $\xrightarrow{\text{electrolysis}}$	(q)	Nitration
(c)	CH_3–Br $\xrightarrow[\text{Dryether}]{\text{Na}}$	(r)	Wurtz reaction
(d)		(s)	Friedal-craft alkylation

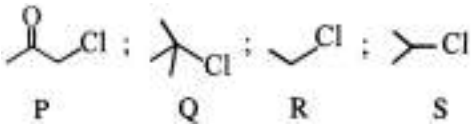
(1) a–p, b–q, c–r, d–s

(2) a–q, b–r, c–p, d–s

(3) a–q, b–p, c–s, d–r

(4) a–q, b–p, c–r, d–s

8) The relative reactivity of following halides towards S_N2 reaction follows the order :-



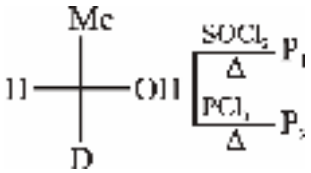
P Q R S

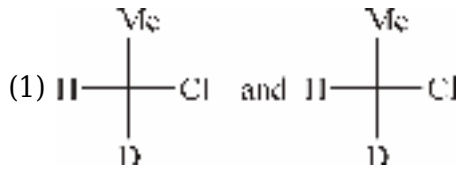
(1) Q > S > R > P

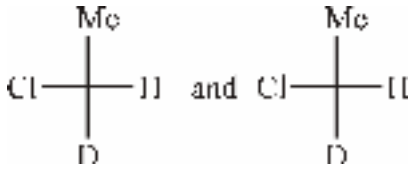
(2) P > S > R > Q

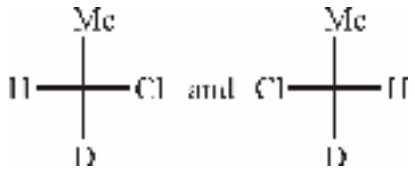
(3) S > R > Q > P

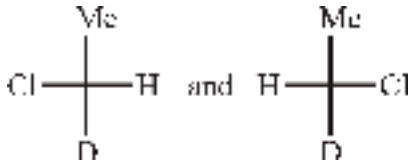
(4) P > R > S > Q

9) Select the correct product from the following:  P_1 and P_2 are respectively :

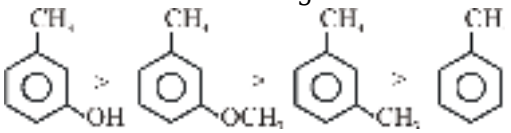
(1) 

(2)  and

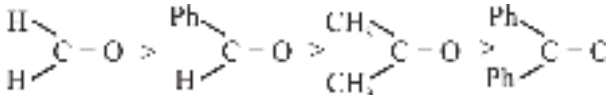
(3) 

(4) 

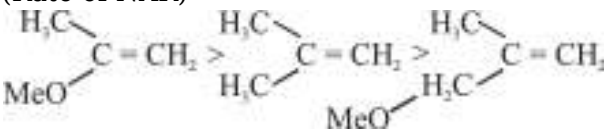
10) Which of the following order is correct :-

(a) 

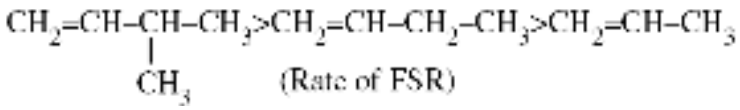
(Rate of ESR)

(b) 

(Rate of NAR)

(c) 

(Rate of EAR)

(d) 

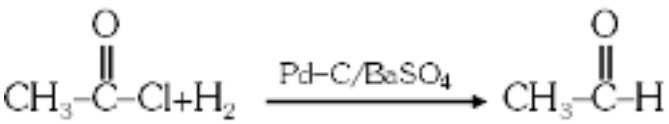
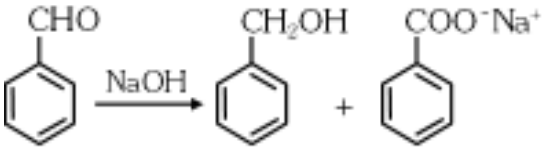
(1) Only a, b, c

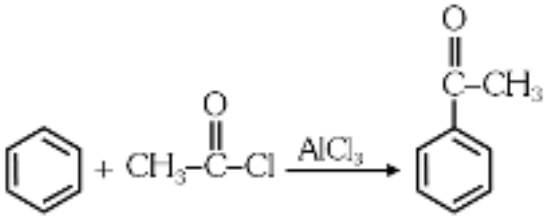
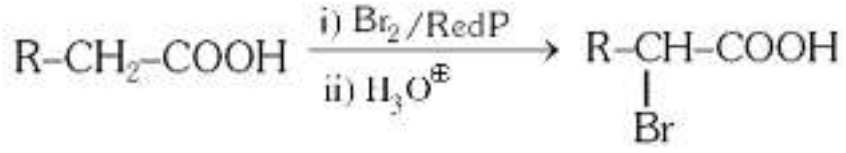
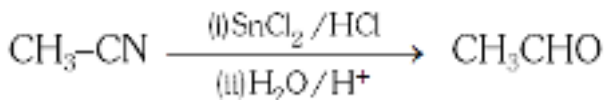
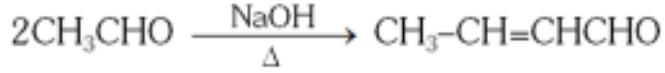
(2) Only b, c

(3) Only b, c, d

(4) a, b, c, d

11) Which of the following column-I (Reaction) is correctly matched with column-II (Reaction name)

	Column I (Reaction)	**Column II (Reaction Name)**
(a)		Stephen's reaction
(b)		Cannizzaro's reaction

(c)		Friedel Crafts alkylation reaction
(d)		HVZ reaction
(e)		Rosenmund's reduction
(f)		Aldol condensation

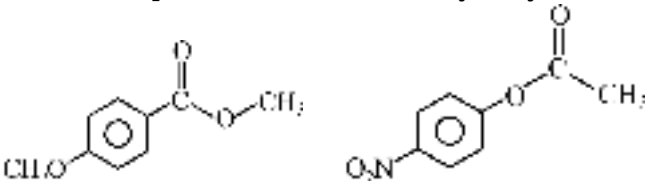
(1) b, c, d, f

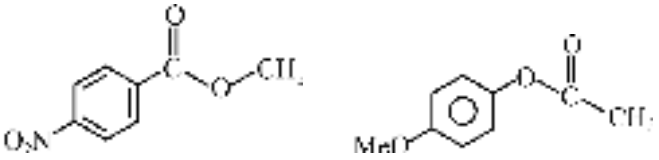
(2) b, d, f

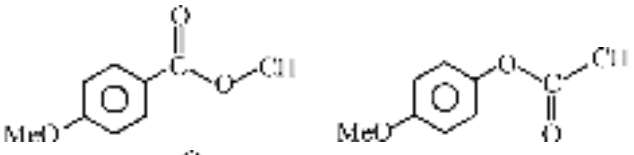
(3) a, b, d, f

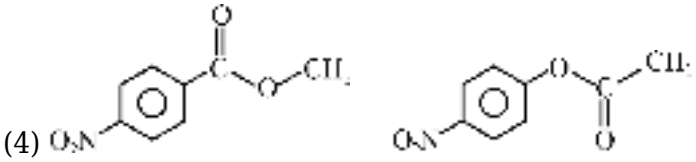
(4) b, c, d, e, f

12) Select the pair in which, rate of hydrolysis is fastest :-

(1) 

(2) 

(3) 

(4) 

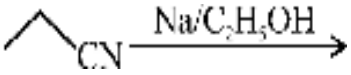
(1) 1

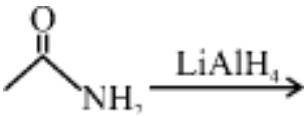
(2) 2

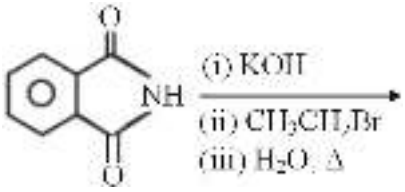
(3) 3

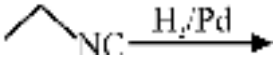
(4) 4

13) Which of the following reaction does not give primary amine as a product ?

(1) 

(2) 

(3) 

(4) 

14)

Match the items given in Column-I with that given in Column-II.

	Column-I		**Column-II**
(a)	Nucleotide	(p)	Linkage in carbohydrates
(b)	Peptide linkage	(q)	Linkage in proteins
(c)	Glycoside linkage	(r)	A sugar and heterocyclic base combination
(d)	Nucleoside	(s)	The monomeric unit in nucleic acids

(1) a → s; b → q; c → p; d → r

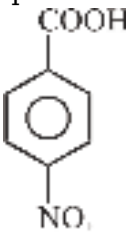
(2) a → q; b → p; c → s; d → r

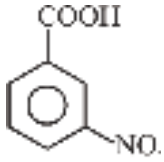
(3) a → p; b → s; c → r; d → q

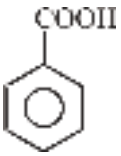
(4) a → s; b → r; c → q; d → p

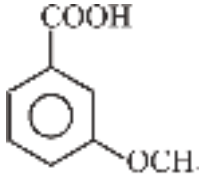
15)

Compare the acidic strength in the following compound?

(a) 

(b) 

(c) 

(d) 

(1) a > d > b > c

(2) a > b > d > c

(3) c > b > a > d

(4) b > a > c > d

16) Which of the following process is exothermic ?

(1) $NaCl_{(s)} \rightarrow Na^{+}_{(g)} + Cl^{-}_{(g)}$

(2) $S_{(g)} + 2e^{-} \rightarrow S^{-2}_{(g)}$

(3) $N^{-}_{(g)} \rightarrow N_{(g)} + e^{-}$

(4) $C^{-}_{(g)} + e^{-} \rightarrow C_{(g)}^{-2}$

17)

Which of the following order is incorrect ?

(1) $Cl_2O_7 > SO_3 > P_4O_{10}$ (acidic strength)

(2) $N > O > P > S$ (IE_1)

(3) $Mn_2O_7 > Mn_2O_3 > MnO_2$ (acidic strength)

(4) $Li^{+}_{(aq)} > Na^{+}_{(aq)} > K^{+}_{(aq)}$ (Radius)

18) Arrange the following statements in order of True(T)/False (F) :-
(a) Among Li^{+}, Be^{+} and B^{+} ions Li^{+} has the highest ionization enthalpy
(b) The negative value of electron gain enthalpy of Cl > F because there is weak electron-electron repulsion in the bigger 3p-subshell of Cl as compared to compact 2p-subshell of F
(c) Formation of S^{-2} and Ar^{-}, both require the absorption of energy
(d) Incorrect order of electron affinity is S > Se > Te > O

(1) TTTF

(2) FFFT

(3) FFTT

(4) TTTT

19) $O^{-}_{(g)} + e^{-} \rightarrow O^{-2}_{(g)}$; $\Delta_{eg}H = a$
$S^{-}_{(g)} + e^{-} \rightarrow S^{-2}_{(g)}$; $\Delta_{eg}H = b$
Values of 'a' and 'b' are respectively :

(1) –141, –300

(2) –300, –141

(3) +800, +640

(4) +640, +800

20) Consider the following order
(a) $PH_3 > NH_3 > AsH_3$ (basic character)
(b) $PH_3 > NH_3 > AsH_3$(boiling point)
(c) $HOCl > HClO_2 > HClO_3 > HClO_4$ (oxidising property)
(d) HF > HCl > HBr > HI (acidic nature)
(e) $H_2O > H_2S > H_2Se$ (bond angle)
(f) $H_2SO_4 > H_3PO_4 > H_2CO_3$ (acidic character)
correct order(s) are :-

(1) a,b,d,e

(2) b,c,e,f

(3) c,e,f

(4) d,e,f

21) Which complex does not show geometrical isomerism but show optical isomerism-

(1) $[M(AA)_2a_2]$

(2) $[M(AA)a_4]$

(3) [Mabcd] tetrahedral

(4) [Mabcd] square planar

22) Match list-I with list-II and select

	List-I		**List-II**
(A)	Zeigler natta	(i)	$Fe_4[Fe(CN)_6]_3$
(B)	Brown ring complex	(ii)	$[Fe(H_2O)_5NO]SO_4$
(C)	Prussian Blue	(iii)	$Al(C_2H_5)_3 + TiCl_4$
(D)	Turnbull Blue	(iv)	$Fe_3[Fe(CN)_6]_2$

Correct code is :-

	A	**B**	**C**	**D**
(1)	(iii)	(ii)	(iv)	(i)
(2)	(iii)	(i)	(iv)	(ii)
(3)	(iii)	(ii)	(i)	(iv)
(4)	(i)	(ii)	(iii)	(iv)

(1) 1

(2) 2

(3) 3

(4) 4

23) The correct order for the wavelength of absorption in the visible region is –

(1) $[Ni(NO_2)_6]^{4-} < [Ni(NH_3)_6]^{2+} < [Ni(H_2O)_6]^{2+}$

(2) $[Ni(NO_2)_6]^{4-} < [Ni(H_2O)_6]^{2+} < [Ni(NH_3)_6]^{2+}$

(3) $[Ni(H_2O)_6]^{4-} < [Ni(NH_3)_6]^{2+} < [Ni(NO_2)_6]^{4-}$

(4) $[Ni(NH_3)_6]^{2+} < [Ni(H_2O)_6]^{2+} < [Ni(NO_2)_6]^{4-}$

24) Select Complex which has all the following characteristics .
(i) Paramagnetic
(ii) No unpaired electrons in t_{2g} orbital
(iii) Show's geometrical isomerism.

(1) $K_4[Fe(NO_2)_2(CN)_4]$

(2) $[Zn(NH_3)_4(en)] Cl_2$

(3) $[Co(en)_3]Cl_3$

(4) $[Ni(NH_3)_2 (en)_2]SO_4$

25) Which of the following compound does not exist :-

(1) VF_5

(2) VCl_5

(3) VI_5

(4) Cr_2O_3

26)

Which of the following arrangements does not represent the correct order of the property stated against it?

(1) $V^{2+} < Cr^{2+} < Mn^{2+} < Fe^{2+}$: paramagnetic behaviour

(2) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$: ionic size

(3) $Co^{3+} < Fe^{3+} < Cr^{3+} < Sc^{+3}$: stability in aqueous solution

(4) Sc < Ti < Cr < Mn : number of oxidation states

27) Which of the following is correct with respect to melting point of a transition element?

(1) V > Cr

(2) Cr > Mn

(3) Mn > Fe

(4) Ti > V

28) Pair of ions which gives blue colour in aqueous state :
V^{+4}, Ni^{+2}, Ti^{+3}, Co^{+2}, Fe^{+3}, Cu^{+2}

(1) V^{+4}, Cu^{+2}

(2) Co^{+2}, Ni^{+2}

(3) Fe^{+3}, Ti^{+3}

(4) Cr^{+3}, Zn^{+2}

29) Which of the following oxy acids of sulphur contains S-S bond ?

I	$H_2S_2O_6$	II	$H_2S_2O_5$
III	$H_2S_2O_4$	IV	$H_2S_2O_8$

(1) I and II

(2) I, II and IV

(3) I and III

(4) I, II and III

30) To avoid the precipitation of hydroxides of Ni^{2+}, Co^{2+}, Zn^{2+} and Mn^{2+} along with those of Fe^{3+}, Al^{3+} and Cr^{3+} the third group solution should be :-

(1) Heated with a few drops of conc. HNO_3

(2) Treated with excess of NH_4Cl

(3) Concentrated

(4) None of these

31) Consider the decomposition of hydrogen peroxide which is catalysed by iodide ion in an alkaline medium :

$$2H_2O_2 \xrightarrow[\text{alkaline medium}]{I^-} 2H_2O + O_2$$

with the following mechanism :

$H_2O_2 + I^- \xrightarrow{\text{I step}} H_2O + IO^-$ (slow)

$H_2O_2 + IO^- \xrightarrow{\text{II step}} H_2O + I^- + O_2$ (fast)

The rate equation for this reaction is found to be :

$$r = \frac{-d[H_2O_2]}{dt} = K[H_2O_2][I^-]$$

which is correct regarding the above reaction :

(A) Reaction is I^{st} order with respect to both H_2O_2 & I^-.

(B) Reaction is elementary reaction.

(C) I^{st} step & IInd step of reaction are Bimolecular elementary reaction.

(D) IO^- is intermediate.

(E) Rate of formation of O_2 is equal to rate of reaction.

(F) II step is RDS.

(1) A, B, C, D, E

(2) A, B, D

(3) A, C, D, E

(4) A, C, D

32) If C(Diamond) → C(graphite) + a kJ mol^{-1} .....(1)

C(Diamond) + $O_{2(g)}$→ $CO_{2(g)}$ + b kJ mol^{-1} .....(2)

C(Graphite) + $O_{2(g)}$ → $CO_{2(g)}$ + c kJ mol^{-1} .....(3)

at constant temp. Then correct relation is :-

(1) –a = b + c

(2) a = b – c

(3) a = c – b

(4) a = b + c

33) The following data were obtained during the first order thermal decomposition of $N_2O_{5(g)}$ at constant volume :

$2N_2O_{5(g)} \rightarrow 2N_2O_{4(g)} + O_{2(g)}$

time (s)	Total pressure (atm)
0	1
100	1.1

find the half life ?

(1) 5 min

(2) 75 min

(3) 300 min

(4) 30 min

34) The following reaction is performed at 298 K

$2NO_{(g)} + O_{2(g)} \rightleftharpoons 2NO_{2(g)}$

The standard free energy of formation of NO is 86600 J mol^{-1} at 298 K. What is the standard free energy of formation of NO_2 at 298 K ? (given equilibrium constant of reaction is 1.6×10^{12})

(1) $86600 + R(298)\ \square n\ (1.6 \times 10^{12})$

(2) $\dfrac{86600}{\ln(1.6 \times 10^{12})} - R(298)$

(3) $0.5\ [2 \times 86600 - R\ (298)\ \square n\ (1.6 \times 10^{12})]$

(4) $R(298)\ \square n\ (1.6 \times 10^{12}) - 86600$

35) At 298 K, 100g of 98% (w/w) H_2SO_4 aq. solution is mixed with 100 g of 49% (w/w) H_2SO_4 aq. solution. What is the molality of H_2SO_4 in the Resultant solution ?

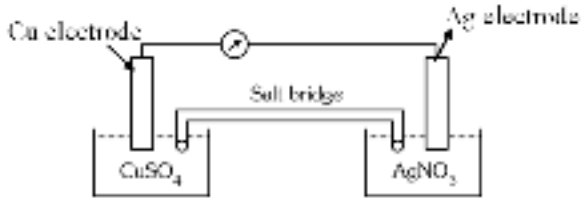
(1) 0.6

(2) 28.3

(3) 41.8

(4) 2.83

36) 1 F charge is passed through 1 L, 1.5 M solution of $CuSO_4$ & 0.1 F charge is passed through 1 L, 0.2 M $AgNO_3$ solution using Pt electrode. After that these solution are used to prepare a galvanic cell as shown below.



The EMF of galvanic cell is :
(assume volume of solution remains same)

[given : $E^{o}_{Cu^{2+}/Cu} = 0.34V,\ E^{o}_{Ag^{+}/Ag} = 0.8V,\ \dfrac{2.303RT}{F} = 0.06$]

(1) 0.2 V

(2) 0.4 V

(3) 0.46 V

(4) 0.26 V

37) In H atom spectrum, (R → Rydberg constant)

(A) The maximum wavelength of Radiation of Lyman series is $\dfrac{4}{3R}$

(B) Balmer series lies in the visible region.

(C) The minimum wavelength of radiation of paschen series is $\dfrac{9}{R}$.

(D) Last line of Lyman has the radiation of maximum frequency.

(E) Total number of spectral lines is $(n_2 - n_1)$

Which is/are correct ?

(1) A, B, C, D

(2) A, B, D

(3) A, B, C

(4) A, B, C, D, E

38) Molecules of Benzoic acid dimerise in benzene. 'W' g of this acid dissolved in 30g of benzene shows a elevation in boiling point equal to 2K. If the % association of acid to form dimer in the solution is 80. Find 'W' ? (K_b = 5 K Kg mol^{-1})

(1) 1.5 g

(2) 1.8 g

(3) 1 g

(4) 2.4 g

39)

SO_2Cl_2 on reaction with excess of water results in to acidic mixture.
$SO_2Cl_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl$
16 moles of NaOH is require for the complete neutralisation of resultant acidic mixture. The number of moles of SO_2Cl_2 used is :-

(1) 16

(2) 4

(3) 8

(4) 2

40) Consider a reaction taking place in closed container at T(K)
$A_{2(g)} + B_{2(g)} \rightleftharpoons 2AB_{(g)}$
2 mol each of $A_{2(g)}$, $B_{2(g)}$ & $AB_{(g)}$ are present at equilibrium. Now 1 mol each of A_2 & B_2 are added to the equilibrium keeping temperature at T(K). The number of moles of A_2, B_2, & AB at new equilibrium respectively will be :

(1) 2.56, 1.62, 2.24

(2) 2.67, 2.67, 2.67

(3) 1.21, 2.24, 1.56

(4) 1.66, 1.66, 1.66

41) Calculate the equilibrium constant of the Cell Reaction of a Cell Represented by :
$Cu_{(s)}|Cu^{2+}_{(aq)}||Ag^{+}_{(aq)}|Ag_{(s)}$
$E^{o}_{Cu/Cu^{+}} = -0.35V, \quad E^{o}_{Cu^{+}/Cu^{2+}} = -0.25V$
$E^{o}_{Ag^{+}/Ag} = 0.8V$

(1) 4×10^{15}

(2) 4×10^{14}

(3) 8×10^{15}

(4) 8×10^{16}

42) Ionisation constant of weak monobasic acid (HA) in terms of λ_M & $\lambda°_M$ is represented by :
(Where λ_M & $\lambda°_M$ are molar conductance at concentration C & at infinite dilution)

(1) $K_a = \frac{C\lambda_M^2}{1-\lambda_M^2}$

(2) $K_a = \frac{C\lambda_M^2}{\lambda_M^o(\lambda_M^o-\lambda_M)}$

(3) $K_a = \frac{C\lambda_M^2}{(\lambda_M^o)^2(\lambda_M^{o2}-\lambda_M^2)}$

(4) $K_a = \frac{C\lambda_M^2}{\lambda_M^o\left(1-\frac{\lambda_M}{\lambda_M^o}\right)}$

43) The sum of number of moles of $KMnO_4$ required for the oxidation of 1 mol each of $FeSO_4$, FeC_2O_4 & $Fe(NO_3)_2$ in acidic medium will be :

(1) 2

(2) 3/5

(3) 4/5

(4) 1

44) Which of the following mixture give a buffer solution of pH = 9.25? (given pK_b of NH_4OH = 4.75)

(1) 0.5 M NH_4OH (0.2 L) + 0.2 M HCl (0.5 L)

(2) 0.2 M NH_4OH (0.5 L) + 0.1 M HCl (0.5 L)

(3) 0.2 M NH_4OH (0.4 L) + 0.1 M HCl (1 L)

(4) 0.4 M NH_4OH (1 L) + 0.1 M HCl (1 L)

45) During Redox titration of $KMnO_4$ v/s oxalic acid which is correct ?

(1) $KMnO_4$ act as self indicator

(2) Starch is used as indicator

(3) HCl is used acid for providing acidic medium

(4) Oxalic acid act as oxidising agent

BIOLOGY

1) Regarding to competition find out the **wrong** statement :-

(1) Unrelated species could compete for same resources

(2) In competition, fitness of one species lowered in presence of other species

(3) Abingdon tortoise become extinct due to competitor starfish

(4) *Balanus* leads to exclusion of *chathamalus* from rocky sea coasts of Scotland

2) Which of the following statements about parasitism are **correct**?
(A) Parasites can be ectoparasites or endoparasites.
(B) All parasites kill their hosts.
(C) Parasites often evolve specialized adaptations to exploit their hosts.
(D) Parasitism always benefits the host in some way.

(1) Only A and C are correct.

(2) Only B and D are correct.

(3) Only A, B, and C are correct.

(4) All statements are correct.

3) Which statements are **correct** ?
(a) In decomposition, complex organic compounds of detritus are converted to CO_2, H_2O and inorganic nutrients.
(b) Species composition and stratification are two main functional features of ecosystem.
(c) Producers in aquatic ecosystem can be higher plants.
(d) Many ecologists regard the entire biosphere as a global ecosystem.

(1) a, b only

(2) b, c only

(3) a, b, c only

(4) a, c, d only

4) Match the following columns correctly.

	I (Component)		**II (Type/Feature)**		**III (Function/Process)**		**IV (Example)**
1.	Primary Producers	A.	Heterotrophs	W.	Conversion of organic matter into inorganic nutrients	i.	Bacteria
2.	Primary consumers	B.	Autotrophs	X.	Traps solar energy by photosynthesis	ii.	Grass
3.	Decomposers	C.	Saprotrophs	Y.	Feed directly on plants	iii.	Deer
4.	Secondary consumers	D.	Carnivores	Z.	Feed on herbivores	iv.	Frog

(1) 1–B–W–ii, 2–A–Z–iii, 3–C–X–i, 4–D–Y–iv

(2) 1–A–Y–ii, 2–B–X–iii, 3–D–W–i, 4–C–Z–iv

(3) 1–B–X–ii, 2–A–Y–iii, 3–C–W–i, 4–D–Z–iv

(4) 1–C–X–i, 2–B–Y–ii, 3–A–W–iii, 4–D–Z–iv

5) The primary consumers in a pond ecosystem are :-

(1) Zooplanktons

(2) Phytoplanktons

(3) Algae

(4) Bacteria

6) Given below are two statements :
Statement-I : Habitat loss and fragmentation is the most important cause driving animals and plants to extinction.
Statement-II : Amazon rain forest is called the lungs of the earth.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Statement I is incorrect, Statement II is correct.

(4) Statement I is correct, Statement II is incorrect.

7) Which of the following is an example of ex-situ conservation ?

(1) Sacred grooves

(2) Botanical gardens and seed banks

(3) National parks

(4) Biosphere reserves.

8) Match **list-I** with **list-II**.

	List-I		**List-II**
A.	The Evil Quartet	I.	Zoological parks
B.	Ex-situ Conservation	II.	Alien Species invasion
C.	*Clarias gariepinus*	III.	Habitat loss and fragmentation.
D.	Quagga	IV.	Extinct

Choose the option with all **correct** matches.

(1) A-III, B-II, C-I, D-IV

(2) A-III, B-I, C-II, D-IV

(3) A-III, B-IV, C-II, D-I

(4) A-III, B-II, C-IV, D-I

9) **Statement-I :** Single DNA-dependent RNA polymerase catalyses transcription of all type of RNA in bacteria.
Statement-II : In bacteria there are three major types of RNA_s: mRNA, t-RNA and rRNA.
Select the correct answer from the options given below :-

(1) Both statement-I and statement-II are correct.

(2) Only statement-I is correct.

(3) Only statement-II is correct.

(4) Both statement-I and statement-II are incorrect.

10) Which of the following is function of m-RNA.

(1) It act as an enzyme.

(2) It provides template for protein synthesis.

(3) It brings the amino acids.

(4) It reads the genetic code.

11) Which of the following statement is not correct ?

(1) Meselson and Stahl proposed a scheme for replication of DNA.

(2) ^{15}N isotope can be separated from ^{14}N only based on densities.

(3) In experiment of Meselson and Stahl DNA extracted from the culture after 40 minutes composed of equal amount of hybrid and light DNA.

(4) The experiment of Taylor proved that DNA in chromosomes also replicate semiconservatively.

12) Regarding HGP methodologies, consider the following interpretations :
I. ESTs focus on identifying all genes expressed as RNA.
II. Sequence Annotation involves sequencing the whole genome including non-coding regions.
III. BAC and YAC are specialized vectors used for cloning DNA fragments in bacteria and yeast.
IV. Automated DNA sequencers used in HGP were based on principles developed by Watson and Crick.
Identify the correct statements :

(1) I, II, III only

(2) II, III, IV only

(3) I and III only

(4) All are correct

13) **Assertion :** In the first phase of translation itself amino acids are activated in the presence of ATP.
Reason : If 2 charged tRNAs are brought close enough. The formation of peptide bond between them would be favoured energetically.

(1) Both Assertion and Reason are correct Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are incorrect.

(3) Both Assertion and Reason are correct Reason is not correct explanation of Assertion.

(4) Assertion is wrong Reason is correct.

14) Find out the correct match from the columns I, II and III :-

Column-I		**Column-II**		**Column-III**	
(I)	Meselson and Stahl	(a)	Bacteriophage	(i)	Proved that DNA replication is semiconservative in bacteria
(II)	Hershey and Chase	(b)	*Vicia faba*	(ii)	Proved that DNA in chromosomes also replicate semiconservatively

(III)	Griffith	(c)	*E.coli*	(iii)	Transformation principle
(IV)	Taylor	(d)	*Streptococcus pneumoniae*	(iv)	Proof that DNA is genetic material

(1) I-c-ii, II-a-iii, III-d-iv, IV-b-i

(2) I-a-i, II-b-iii, III-d-iv, IV-c-ii

(3) I-c-i, II-a-iv, III-d-iii, IV-b-ii

(4) I-b-iv, II-c-ii, III-a-i, IV-d-iii

15) **Assertion:** Male heterogamety is present in grasshopper
Reason: The males have one chromosome less than females in grashopper.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

16) Match Column-I with Column-II and select the correct option from the codes given below.

Column-I		**Column-II**	
(A)	Dihybrid test cross	(i)	9 : 7
(B)	Law of segregation	(ii)	Dihybrid cross
(C)	Law of independent assortment	(iii)	1 : 1 : 1 : 1
(D)	ABO blood group in man	(iv)	Purity of gametes
		(v)	Multiple allelism

(1) A-(iii), B-(iv), C-(ii), D-(v)

(2) A-(i), B-(iv), C-(ii), D-(v)

(3) A-(iii), B-(ii), C-(iv), D-(v)

(4) A-(ii), B-(v), C-(iii), D-(i)

17) Match list-I with list-II :

List-I		**List-II**	
A.	$I^A i$, rr	I	O^- blood group
B.	$I^A I^B$, Rr	II	A^- blood group
C.	i i, rr	III	B^+ blood group
D.	$I^B I^B$, Rr	IV	AB^+ blood group

Choose the correct answer from options given below :-

(1) A-II, B-IV, C-III, D-I

(2) A-II, B-IV, C-I, D-III

(3) A-II, B-I, C-IV, D-III

(4) A-III, B-IV, C-I, D-II

18)

	Column-A		Column-B
(A)	Human sex-determination	(i)	Haplo-diploid system
(B)	Birds	(ii)	xo-xx system
(C)	Grasshoppers	(iii)	zw-zz system female heterogenetic
(D)	Honey bees	(iv)	xx-xy system male heterogametic

(1) A-iv, B-iii, C-ii, D-i

(2) A-i, B-iii, C-ii, D-iv

(3) A-iii, B-i, C-ii, D-i

(4) A-iv, B-ii, C-iii, D-i

19) Scientist who confirmed the release of a volatile substance from ripened oranges that hastened the ripening of stored unripened bananas.

(1) E. Kurosawa

(2) F.W. Went

(3) H.H. Cousins

(4) Miller

20) **Assertion :** The plant growth regulators are small, simple molecules of diverse chemical composition.
Reason : The major groups of plant growth regulators, such as auxins, gibberellins, cytokinins, abscisic acid and ethylene, were identified as a result of accidental observations during various experiments.
Options :-

(1) Both assertion and reason are true but reason is not the correct explanation of assertion.

(2) Both assertion and reason are true and reason is the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

21) Consider the following statements :
(A) Plant Growth Regulators (PGRs) act only as promoters.
(B) PGRs are intrinsic intercellular chemical factors.
(C) A single PGR can produce diverse physiological effects.
(D) Different PGRs may manifest similar effects.
Which is/are **correct**?

(1) A and B

(2) A, C, D

(3) B, C, D

(4) Only C

22) Which statement is **correct** ?

(1) The passage of electrons through the F_0 channel is coupled to the catalytic site of the F_1 component for the production of ATP.

(2) During the conversion of succinyl-CoA to succinic acid a molecule of $FADH_2$ is synthesized.

(3) In mitochondria, complete oxidation of pyruvic acid results in release of three CO_2 and 4 molecules of NADH + H^+; one of $FADH_2$ have been synthesized.

(4) The metabolic pathway in respiration through which the protons are transfered from H_2O to NAD^+, is called the ETS.

23) How many of the following statements are **correct** ?
(A) The respiratory pathway is an amphibolic pathway as it involves both anabolism and catabolism.
(B) The respiratory quotient depends upon the type of respiratory substance used during respiration.
(C) NADH + H^+ and $FADH_2$ are generated in glycolysis.
(D) During electron transport in the inner membrane of mitochondria, synthesis of ATP is known as photophosphorylation.
(E) Fermentation takes place under anaerobic conditions in many prokaryotes, unicellular eukaryotes and in germinating seeds.

(1) 3

(2) 4

(3) 5

(4) 2

24) Cornelius van Niel's studies on purple and green bacteria demonstrated that photosynthesis is essentially a light-dependent reaction in which:

(1) Oxygen get reduced to water

(2) Carbon dioxide is reduced to carbohydrates by hydrogen from a suitable oxidizable compound.

(3) Sulphur is released as an reduction product.

(4) Light energy is converted to heat energy.

25) Given below are two statements : One is labelled as **Assertion** and other is labelled as **Reason**.
Assertion : In C_4 plants, the bundle sheath cells may form several layers around the vascular bundles, they are characterised by having a large number of chloroplasts, thick walls impervious to gaseous exchange & no intercellular spaces.
Reason : In C_4 plants the mesophyll cells lack RuBisCO enzyme but have PEP carboxylase enzyme.
In the light of the above given statements, choose the correct answer from the options given below :

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

26) Which of the following secondary metabolites are categorised as drugs ?

(1) Abrin and ricin

(2) Vinblastin and curcumin
(3) Arthocyanins
(4) Gums and cellulose

27) Match **List I** with **List II**.

	List I (Element)		**List II (% weight of human body)**
(A)	Ca	(i)	1.5
(B)	C	(ii)	18.5
(C)	O	(iii)	65.0
(D)	N	(iv)	3.3

Options :-
(1) (A) – (ii), (B) – (iii), (C) – (iv), (D) – (i)
(2) (A) – (ii), (B) – (iv), (C) – (iii), (D) – (i)
(3) (A) – (i), (B) – (ii), (C) – (iv), (D) – (iii)
(4) (A) – (i), (B) – (ii), (C) – (iii), (D) – (iv)

28) Prosthetic groups differ from Co-enzymes in that :-

(1) they require metal ion for their activity.
(2) They are tightly bound to apoenzyme.
(3) their association with apoenzyme is transient.
(4) they can serve as Co-factors.

29) Which of the following statement is correct ?

(1) Competitive inhibitors are often used in the control of bacterial pathogens.
(2) Enzymes generally function in a wide range of temperature and pH.
(3) Succinate is competitive inhibitor of succinate dehydrogenase enzyme.
(4) Change in temperature and pH can alter only the primary structure of the protein in an enzyme.

30) Which layer is not observed in the transverse section of dicot root ?

(1) Epidermis
(2) Cortex
(3) Endodermis
(4) Hypodermis

31) Which one is not a function of root ?

(1) Absorption of water and minerals from soil.
(2) Providing a proper anchorage to the plant parts

(3) It conducts water, minerals and photosynthates

(4) Storing reserve food material

32) Select incorrect statement ?

(1) Ovary is inferior in flowers of guava and cucumer.

(2) In pea flower, vexillary aestivation is found in petals.

(3) Polyadelphous condition of stamens is found in citrus

(4) Basal placentation is found in primrose.

33) Study the column-I, II and III and identify the correct match.

Column-I		**Column-II**		**Column-III**	
(A)		(P)	Vexillary	(i)	*Calotropis*
(B)		(Q)	Valvate	(ii)	China rose
(C)		(R)	Imbricate	(iii)	Cassia
(D)		(S)	Twisted	(iv)	Pea

(1) Only A is correct

(2) Only B is correct

(3) Only C is correct

(4) Only A and D is correct

34) Given below are two statements :
Statement-I : As compared to the dicot root which have fewer xylem bundles, there are usually more than six (polyarchy) xylem bundles in the monocot root.
Statement-II : In grasses, certain abaxial epidermal cells along the veins modified themselves into bulliform cells.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both statement-I and statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Both statement-I and statement-II are correct.

(4) Statement-I is incorrect but statement-II is correct.

35) Which of the following division occurs in microspore mother cell during microspore formation?

(1) Mitosis

(2) Meiosis

(3) Meiosis followed by mitosis

(4) Endomitosis

36) It root cell of flowering plant contain 14 chromosomes then what will be number of chromosome in aleurone layer of seed of same plant.

(1) 14

(2) 7

(3) 21

(4) 28

37) Match the column-I with column-II :-

Column-I		**Column-II**	
(A)	Autogamy	(i)	Different plant but same species
(B)	Geitonogamy	(ii)	Same plant but different flower
(C)	Xenogamy	(iii)	Same flower

(1) (A)-(i), (B)-(ii), (C)-(iii)

(2) (A)-(ii), (B)-(i), (C)-(iii)

(3) (A)-(iii), (B)-(i), (C)-(ii)

(4) (A)-(iii), (B)-(ii), (C)-(i)

38) Which of the following statements is correct ?

(1) Each taxonomic category represents a rank and is commonly termed as a taxon

(2) All organisms have species as the highest category

(3) Taxonomic categories are merely morphological aggregates

(4) Classification is a single step process

39) Which is the cause of tetanus and cholera ?

(1) Virus

(2) Bacteria

(3) Fungi

(4) Algae

40) Match the column-I with column-II :-

Column-I	**Column-II**
A. Prions	i. Pollution indicators
B. Viroids	ii. Lacked the protein coat

C. Lichens	iii. Mosaic disease in Tobacco
D. Virus	iv. Neurological disease

Choose the correct answer from the options given below :-

(1) A-ii, B-iv, C-i, D-iii

(2) A-iv, B-ii, C-i, D-iii

(3) A-i, B-ii, C-iv, D-iii

(4) A-iv, B-ii, C-iii, D-i

41) (I) Bacteria reproduce mainly by ____A____ .
(II) Pigments of Eugleoids are identical to ____B____ .
(III) Example of ciliated protozoan is ____C____ .
(IV) *Puccinia* causes ____D____ .
Select correct option for A, B, C, D and fill in the plants.

(1) A-Mitosis, B-Chrysophytes, C-*Plasmodium*, D-Smut

(2) A-Meiosis, B-Lower plants, C-*Amoeba*, D-Smut

(3) A-Mitosis, B-Higher plants, C-*Amoeba*, D-Rust

(4) A-Fission, B-Higher plants, C-*Paramoecium*, D-Rust

42) How many species are in the list given below :
Mangifera indica, Solanum tuberosum, Panthera leo, Tigris, Melongena, Nigrum, Homo sapiens.

(1) Three

(2) Eight

(3) Four

(4) Seven

43) Which option is not related with phaeophyceae?

(1) holdfast

(2) Algin

(3) Fucoxanthin

(4) Floridean starch

44) Read the following statement and choose the correct option :-
Statement-I : In gymnosperms pollen tube carrying the male gametes grows towards archegonia in the ovules and discharge their male gametes near the mouth of the archegonia.
Statement-II : Double fertilisation and triple fusion is unique event shown by angiosperms.

(1) Both statements are correct.

(2) Both statements are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

45) Identify the figure A & B and related characters :

	 **(A)**	 **(B)**
(1)	*Cycas* - Pinnate leaves	Dicotyledon - ovules are developed in flower
(2)	Monocotyledon - Pollen grains	Dicotyledon - Ovules are naked
(3)	*Pinus* - coralloid roots	*Ginkgo* - Naked seeds
(4)	*Pinus* - The male and female cones borne on same tree	Monocotyledon - Seeds are enclosed in fruits

(1) 1

(2) 2

(3) 3

(4) 4

46) Read the following statements :-
(a) Members of phylum porifera are known as sea walnuts or combjellies.
(b) Flatworms are bilaterally symmetrical, triploblastic and coelomate animals.
(c) *Planaria* possess high regeneration capacity.
(d) Aquatic annelids like Nereis possess lateral appendages parapodia.
(e) Most distinctive feature of echinoderms is the presence of water transport system.
Find the correct statements.

(1) a, b, c and e only

(2) a, b, d and e only

(3) c and d only

(4) c, d and e only

47) Match list I with List II

	List I		**List II**
(A)	Ascaris	(i)	Diploblastic
(B)	Tapeworm	(ii)	Hemichordata
(C)	Stomochord	(iii)	Pseudosegmented
(D)	Hydra	(iv)	Tube within tube body plan

(1) (A) – (i), (B) – (ii), C – (iii), D – (iv)

(2) (A) – (iv), (B) – (iii), C – (ii), D – (i)

(3) (A) – (iii), (B) – (i), C – (iv), D – (ii)

(4) (A) – (ii), (B) – (iv), C – (i), D – (iii)

48) **Assertion :-** Amphibia are poikoilothermal animals and they cannot regulate their body temperature.

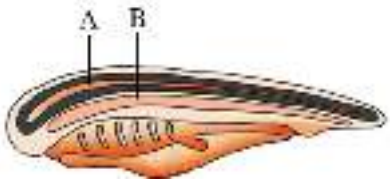
Reason :- In amphibia, body is divided into head and trunk. Tail may be present in some.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Assertion is False but Reason is True.

49) The largest and the second largest phylum of the animal kingdom are __A__ and __B__ respectively :

(1) A-Arthropoda, B-Echinodermata
(2) A-Echinodermata, B-Mollusca
(3) A-Mollusca, B-Vertebrata
(4) A-Arthropoda, B-Mollusca

50) Identify A and B :



(1) A-Nervecord ; B-Stomochord
(2) A-Notochord ; B-Stomochord
(3) A-Notochord ; B-Nervecord
(4) A-Nervecord ; B-Notochord

51) Find out the incorrect statement about the cockroach :-

(1) Their unfertilised eggs are encased in capsule called oothecae.
(2) In male external genitalia are represented by male gonapophysis
(3) In male sperms are stored in the seminal vesicles.
(4) In male, the ejaculatory duct opens into male gonopore situated ventral to anus.

52) Given below are two statements.
Statement-I :- Matrix of cartilage is solid & pliable.
Statement-II :- Matrix is solid so blood circulation is absent in the matrix of cartilage.

(1) Both statement I and statement II are incorrect.
(2) Statement I is correct but statement II is incorrect.
(3) Statement I is incorrect but statement II is correct
(4) Both statement I and statement II are correct.

53) Which one of the following ducts open in the bidders canal of frog ?

(1) Vasa efferentia
(2) Ureters
(3) Vasa deferentia

(4) Urinogenital duct

54) Read column-I, II & III and identify their correct match :-

	Column-I		**Column-II**		**Column-III**
(A)	Thyroxine	(P)	Ca^{+2} Level in blood	(i)	Low intelligence quotient
(B)	PTH	(Q)	Inhibit cellular uptake of amino acid	(ii)	Excretion of K^{+} and phosphate ions.
(C)	Cortisol	(R)	Red blood cells formation	(iii)	Hyper calcemia
(D)	Aldosterone	(S)	Maintenance of electrolytes	(iv)	Anti-inflammatory

(1) A-R-i, B-P-iii, C-Q-iv, D-S-ii

(2) B-P-iii, A-Q-i, C-R-iv, D-S-ii

(3) A-P-i, B-Q-ii, C-R-iii, D-S-iv

(4) A-R-i, B-P-ii, C-Q-iii, D-S-iv

55) Which one of the following statements is correct about thyroid gland ?

(1) Both the lobes are interconnected with a thick flap of connective tissue called isthmus.

(2) Iodine is essential for the normal rate of thyroxine hormone synthesis in the thyroid.

(3) Deficiency of iodine in our diet results in hyperthyroidism and enlargement of the thyroid gland, commonly called goitre.

(4) Exopthalmic goitre is a form of hypothyroidism, characterised by enlargement of the thyroid gland, protrusion of the eye balls, also called Grave's disease.

56) Hormone from adenohypophysis which stimulates the synthesis and secretion of steroid hormones from the adrenal cortex is

(1) Glucocorticoid

(2) ACTH

(3) Mineralocorticoid.

(4) TSH

57) **Assertion :** Pons containing fibre tracts that interconnect different regions of brain.
Reason : Pons is the major co-ordinating centre for sensory and motor signalling.

(1) Assertion and Reason both are false.

(2) Assertion is true but the reason is false

(3) Assertion is false but the reason is true

(4) Assertion and Reason both are true.

58) Which of the following type of neurons are usually found in embryonic stage ?

(1) Multipolar

(2) Bipolar

(3) Unipolar

(4) Pseudounipolar

59) Match the column and choose the correct option :-

	Column-I	**Column-II**
(1)	Fibrous joint	Adjacent vertebrae
(2)	Pelvic girdle	Total three girdle bones in number
(3)	Tetany	Accumulation of uric acid crystals
(4)	Osteoporosis	Decreased levels of estrogen

(1) 1

(2) 2

(3) 3

(4) 4

60) **Statement-I :** During the muscles contraction myosin head binds to the unexposed active site on actin to form a cross bridge.
Statement-II : ATP hydrolysis is required for the breakdown of cross bridge.

(1) Both statement-I and statement-II are correct

(2) Only statement-I is correct

(3) Only statement-II is correct

(4) Both statement-I and statement-II are incorrect

61) Which one of the following statements is incorrect ?

(1) The cardiac events that occur from the beginning of one heart beat to beginning of the next are called cardiac cycle.

(2) Cardiac cycle consists of systole and diastole of both the atria and ventricles.

(3) The heart beats 72 times per minute, i.e., only one cardiac cycle is performed per minute.

(4) During a cardiac cycle, each ventricle pumps out approximately 70 ml of blood which is called the stroke volume.

62) Destruction of RBC occurs in which organ :-

(1) Heart

(2) Lungs

(3) Kidney

(4) Spleen

63) The action potential causes the atria and then the ventricles to undergo contraction (systole) followed by their relaxation (diastole). The systole forces the blood to move from the atria to the ventricels and then to the palmonary artery and the aorta.
Statement-I : The SAN generates an action potential which stimulates both the atria to undergo a simultaneous contraction.
Statement-II : Ventricular pressure increases, allowing the blood transfer from ventricles to atria.

(1) Both statement I and II are correct
(2) Statement I is correct & II is incorrect
(3) Statement I is incorrect & II is correct
(4) Both statement I and II are incorrect

64)

Match the column-I with column-II and choose correct answer.

	Column-I		**Column-II**
A	DCT	i	Filtration of blood
B	Counter current mechanism	ii	Special sensitive region
C	JGA	iii	Maintain concentration gradient of medullary interstitium
D	Malpighian body	iv	Site of conditional reabsorption of Na^+ and water

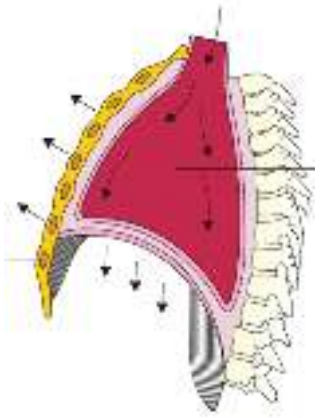
(1) A-ii, B-i, C-iii, D-iv
(2) A-iv, B-ii, C-iii, D-i
(3) A-iv, B-iii, C-ii, D-i
(4) A-ii, B-iii, C-i, D-iv

65) Which of the following structure helps in elimination of lactic acid?

(1) Liver
(2) Sweat glands
(3) Sebaceous gland
(4) Lungs

66)

Identify the process and muscles involved in the given diagram ?



(1) Expiration - contraction of diaphragm and E.I.C.M.

(2) Expiration - Relaxation of diaphragm and E.I.C.M.

(3) Inspiration - Contraction of diaphragm and I.I.C.M.

(4) Inspiration - Contraction of diaphragm and E.I.C.M.

67) Which of the following layers is NOT part of the diffusion membrane involved in gas exchange at the alveoli ?

(1) Squamous epithelium of aveoli

(2) Endothelium of alveolar capillaries

(3) Basement substance

(4) Bronchial epithelium

68) The female is experiencing luteal phase defect (LPD), which leads to decreased secretion of the corpus luteum. As a result, she is unable to conceive, and has had several spontaneous abortions. As an obstetrician what advice would you like to give to this woman so that she can conceive properly?

(1) She should take the help of a surrogate mother for the development of the fetus

(2) Estrogen supplementation

(3) Progesterone supplementation

(4) She cannot conceive, so legal adoption is best option.

69) How many of the following are correct ?
(A) Sperm motility essential for fertilisation.
(B) Human male ejaculate at a time → 200-300 million sperm.
(C) Semen ⇒ seminal plasma along with sperms.
(D) Female sex accessory duct & glands → Maintained by Androgens.
(E) Acrosomal enzymes help in fertilisation of ovum.
Choose correct option.

(1) One

(2) Three

(3) Four

(4) Two

70) What hormone is typically injected by doctors to induce delivery ?

(1) Estrogen

(2) Progesteron

(3) Oxytocin

(4) Testosterone

71) Identify the given figure 'X' and how it works as an ideal contraceptive for the females who want to delay pregnancy and/or space children?



Fig. X

(1) LNG20 : Phagocytosis of sperms within the uterus

(2) Cu-7 : Make the uterus unsuitable for implantation and cervix hostile to the sperm

(3) Lippes loop : Suppress sperm motility and fertilising capacity of sperms

(4) Copper-T : Released Cu-ions which suppress sperm motility and fertilising capacity.

72) Arrange the following events in a correct sequence occur during life cycle of malarial parasite.
(a) Parasite reproduces asexually in RBCs, bursting the RBCs and causing symptoms.
(b) Entry of parasite in liver cells where it reproduces asexually.
(c) Fertilisztion in gut of mosquito.
(d) Sexual stages (gametocytes) formed in RBCs.
(e) Infective stage (sporozoites) migrate to salivary glands of mosquito.

(1) d, a, b, e, c

(2) b, a, d, c, e

(3) a, b, c, d, e

(4) e, a, d, b, c

73) **Assertion (A) :** Smoking is detrimental to health due to the harmful chemicals in tobacco smoke.
Reason (R) : Nicotine stimulate adrenal gland to release adrenaline and nor-adrenaline which causes high blood pressure and lung cancer.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

74) Read the following options carefully :
(a) The use of drugs like antihistamine, adrenalin and steroids quickly reduce the symptoms of allergy.

(b) The lymph nodes are large solid structures located at different point along the lymphatic system.
(c) The spleen is a large lobed organ. It mainly contains lymphocytes and phagocytes
(d) Active immunity in slow and takes time to give its full effective response.
How many statements are correct ?

(1) 1
(2) 2
(3) 3
(4) 4

75) The exaggerated response of the immune system to certain antigens present in the environment is called :-

(1) Primary response
(2) Secondary response
(3) Allergy
(4) Auto immune disease

76) Which activates immune system and helps in destroying the tumor ?

(1) B-lymphocytes
(2) Cyclosporin
(3) α-interferon
(4) Steroids

77) **Statement -I** :- Fitness is the end result of the ability to adapt and get selected by nature.
Statement -II :- When we describe the story of life we describe evolution as a process.

(1) Only statement I is incorrect.
(2) Only statement II is incorrect.
(3) Both statement are incorrect.
(4) Both statements are correct.

78) Population genetics studies how genetic composition changes within populations. The Hardy-Weinberg principle. States that in the absence of evolutionary forces like mutation, natural selection, migration etc. The allele frequencies in a population remain constant from one generation to the next.
Statement-I : The Hardy-Weinberg principle provides a mathematical model that describe a population in which allele frequencies remain stable overtime.
Statement-II : Any deviation from the equilibrium conditions indicates that evolutionary forces are acting on the population leading to changes in gene frequencies.

(1) Both Statement I and Statement II are true.
(2) Both Statement I and Statement II are false.
(3) Statement I is true but Statement II is false.
(4) Statement I is false but Statement II is true.

79) Eyes of the octopus and of mammals are the example of :-

(1) Analogous organ
(2) Homologous organ
(3) Vestigial organ
(4) Parellel organ

80) Which of the following enzyme is not involved in the isolation at genetic material DNA from a plant cell

(1) Cellulase
(2) Pectinase
(3) Lysozyme
(4) Protease

81) Correctly match the column I and II

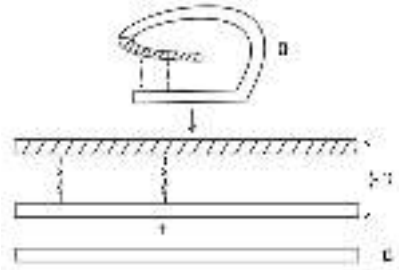
	Column-I		**Column-II**
a	Restriction endonuclease	i	Insecticidal
b	Gel electrophoresis	ii	DNA amplification
c	Cry protein	iii	Separation of DNA fragments
d	PCR	iv	Molecular scissors

(1) a-iii, b-ii, c-iv, d-i
(2) a-i, b-iii, c-iv, d-ii
(3) a-ii, b-iii, c-i, d-iv
(4) a-iv, b-iii, c-i, d-ii

82) Which statement is incorrect :

(1) *Salmonella typhimurium* has plasmid DNA as extrachromosomal DNA.
(2) Each restriction endonuclease functions by inspecting the length of a DNA sequence.
(3) Hind II a palindromic sequence in a DNA stretch.
(4) Sticky ends and blunt ends are equally useful in DNA recombination technology.

83) The given diagram is maturation of insulin, select the correct name labelled A, B & C.



(1) a-A peptide, b-B peptide, c-C peptide
(2) a-proinsulin, b-peptide, c-free C-peptide

(3) a-proinsulin, b- A peptide, c-free C-peptide

(4) a-pro insulin, b-C-peptide, c-B peptide

84) Match the following columns and select the correct option.

Column-I		**Column-II**	
(A)	Bt cotton	(i)	Gene therapy
(B)	Adenosine deaminase	(ii)	Cellular defence
(C)	RNAi	(iii)	Detection of HIV infection
(D)	PCR	(iv)	*Bacillus thuringiensis*

(1) (A) – i, (B) – ii, (C) – iii, (D) – iv

(2) (A) – iv, (B) – i, (C) – ii, (D) – iii

(3) (A) – iii, (B) – ii, (C) – i, (D) – iv

(4) (A) – ii, (B) – iii, (C) – iv, (D) – i

85) **Statement 1 :** GEAC make decisions regarding the validity of GM research and safety of introducing GM organisms for public services.
Statement 2 : Many transgenic animals are designed to increase our understanding of how genes contribute to the development of diseases.

(1) Both Statement 1 and 2 are incorrect.

(2) Both Statement 1 and 2 are correct.

(3) Statement 1 is correct but Statement 2 is incorrect.

(4) Statement 1 is incorrect but Statement 2 is correct.

86) Packaging of macromolecules which has to be delivered outside from cell, occur in :-

(1) Golgi body

(2) Rough Endoplasmic reticulum

(3) Smooth endoplasmic reticulum

(4) Vacoule

87) The stroma of chloroplast does not contain :-

(1) Enzymes for the synthesis of carbohydrates and proteins

(2) Small, double stranded, linear DNA

(3) Ribosomes

(4) ATP and NADPH + H^+

88) Which of the following statement is incorrect ?

(1) All cells arise from pre-existing cells

(2) In both prokaryotic and eukaryotic cells, a semifluid matrix called cytoplasm occupies the volume of the cell

(3) Cell differ greatly in size, shape and activities

(4) Anything less than a complete structure of a cell, ensures independent living

89) Which of the following statement is true about meiosis?
(a) It involves two sequential cycle of nuclear and cell division called meiosis I and II but only a single cycle of DNA replication.
(b) It involves pairing of homologous chromosome and recombination between them.
(c) Four haploid cells are formed at the end of meiosis.

(1) Only (a) and (b) are correct

(2) Only (c) is correct

(3) Only (a) and (c) are correct

(4) All are correct

90) Which of the following statements correct regarding cell cycle & cell division ?
(A) Morphology of chromosomes is best studied during anaphase.
(B) Chromosomes lose their individuality during prophase.
(C) Cells are metabolically active in quiescent stage.
(D) Assembly of mitotic apparatus begins at prophase.
(E) Animal cell cytokinesis is done by furrow formation.

(1) A & B

(2) C, D & E only

(3) D & E

(4) A, C, D, & E

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	1	1	1	3	4	4	4	3	1	4	1	1	2	4	1	1	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	2	1	2	3	3	2	1	2	3	1	1	4	1	3	2	1	1	1	2
Q.	41	42	43	44	45															
A.	1	1	2	4	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	3	1	4	2	4	4	3	4	2	4	4	1	2	3	3	1	3	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	1	4	3	1	2	1	4	2	3	2	1	3	2	2	1	4	2	2
Q.	86	87	88	89	90															
A.	4	2	4	2	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	1	4	3	1	1	2	2	1	2	1	1	3	3	1	1	2	1	3	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	1	2	4	2	4	2	1	4	3	4	3	2	2	3	4	1	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	4	1	4	3	2	2	4	4	1	2	1	1	2	2	2	3	4	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	4	2	3	2	4	4	3	3	3	4	2	3	2	3	3	2	1	1	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	3	2	2	1	2	4	4	2										

SOLUTIONS

PHYSICS

1) NCERT-XI (Part-I), Pg. # 27

$PV^{3/2} = K$

$P = KV^{-3/2}$

$$\frac{\Delta P}{P} = -\frac{3}{2}\frac{\Delta V}{V}$$

$$\frac{\Delta P}{P} \times 100 = -\frac{3}{2}\left(\frac{\Delta V}{V} \times 100\right)$$

$$= -\frac{3}{2} \times -0.5$$

$= 0.75\%$

2) $$S = \int_0^t |v|\,dt = \int_0^2 |v|\,dt + \int_2^4 |v|\,dt$$

$$= \int_0^2 (4 - (2t))\,dt + \int_2^4 (3t - 6)\,dt = 10m$$

3) Horizontal distance, $d = (u \cos \alpha)t \Rightarrow u = \frac{d}{t \cos \alpha}$

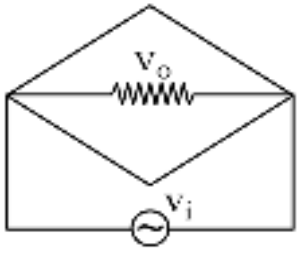
Vertical displacement, $-h = (u \sin \alpha)t - \frac{1}{2}gt^2$.

Substitute u ;

$$-h = \left(\frac{d}{t\cos\alpha}\sin\alpha\right) t - \frac{1}{2}gt^2 \Rightarrow -h = d\tan\alpha - \frac{1}{2}gt^2$$

$$t = \sqrt{\frac{2(h + d\tan\alpha)}{g}}$$

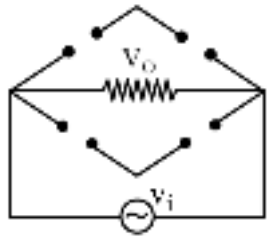
4) During ⊕ve cycle, all diode forward biased



0

V = 0

During ⊖ve cycle, all diode reverse biased



5) $Y = \overline{(A + \bar{B}).(\bar{A} . \bar{B})}$

$= (\bar{A} . \bar{B}).(\bar{\bar{A}} + \bar{\bar{B}})$

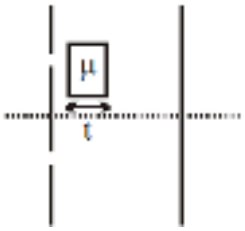
$= (\bar{A} . \bar{B}).(A + B)$

$= \bar{A} . \bar{B} . A + \bar{A} . \bar{B} . B$

$= 0 + 0 = 0$

6) Wave front are ⊥ to rays

7)



$\Delta X = (\mu - 1)t = 1.\lambda$

for one maximum shift

$$\boxed{t = \frac{\lambda}{\mu - 1}}$$

8)

Stopping potential is minimum negative potential of anode w.r.t. cathode at which photocurrent become zero.

9)

All given four statements are correct.

10) $v = \sqrt{2gh}$ = same For both

$\lambda = \frac{h}{mv}$

$$\frac{\lambda_1}{\lambda_2} = \frac{m_2}{m_1} = \frac{4}{2} = \frac{2}{1}$$

11) $V = \frac{150}{[\lambda_e(\text{in Å})]^2}$ volt

$$V = \frac{150}{(0.01)^2} = 150 \times 10^4 \text{ volt}$$

for x rays, $\lambda_{min} = \frac{12400}{V}$ Å

$= \frac{12400}{150 \times 10^4}$ Å

$= 8.26 \times 10^{-3}$ Å

12) Distance between lenses = $|V_0| + |u_e|$

using lens formula for objective,

$|V_0| = 6$ cm

using lens formula for eye piece, $|u_e| = 5$ cm

∴ Distance between two lenses = 6 + 5 = 11 cm

13) $\sin i = \frac{f}{R} = \frac{1}{2}$; i = 30°

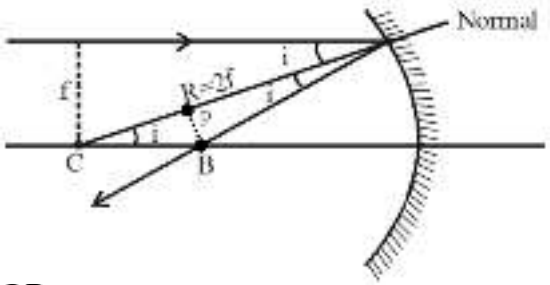
Not a paraxial ray

In ΔCPB

$\cos i = \frac{CP}{CB}$

$\cos i = \frac{(R/2)}{CB}$

$CB = \frac{R}{2\cos i} = \frac{f}{(\sqrt{3}/2)} = f\left(\frac{2}{\sqrt{3}}\right)$



$\frac{CB}{f} = \frac{2}{\sqrt{3}}$

14) $\frac{h}{\sqrt{\mu^2 - 1}} = r$

$r = \frac{4}{\sqrt{\frac{25}{9} - 1}} \Rightarrow r = \frac{4 \times 3}{4}$

r = 3 m

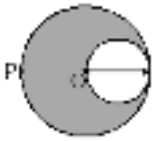
∴ d = 2 × 3 = 6 m

15) $a = \frac{f}{m} = \frac{Bi\ell}{m}$

$= \frac{2 \times 2 \times 3 \times 10^{-2}}{10 \times 10^{-3}} = 12$ ms^{-2}

here effective length = AC = 3 cm

16) If cavity is filled up then $B_1 = \frac{\mu_0 \times I}{2\pi a}$



$B_1 = \frac{\mu_0 \times J \times (\pi a^2)}{2\pi a}$

Magnetic field int. due to smaller cylinder of radius 'a/2' $B_2 = \frac{\mu_0 \times I}{2\pi r}$

$= \frac{\mu_0 \times J \times (\pi (a/2)^2)}{2\pi \left(\frac{3a}{2}\right)}$

$B_{Net} = B_1 - B_2$

17)

$R = \frac{125}{12.5} = 10\ \Omega$

$X_L = \omega L = 2\pi nL = \frac{V}{I} = \frac{125}{10} = 12.5$

$\therefore\ 2\pi nL = 12.5$

or $2\pi L = \frac{12.5}{50} = 0.25$

$\Rightarrow\ X_L = 2\pi L \times n = 0.25 \times 40 = 10\ \Omega$

Impedance of the circuit,

$Z = \sqrt{R^2 + X_L^2} = 10\sqrt{2}$ ohm

$\Rightarrow$ Current $\frac{100\sqrt{2}}{10\sqrt{2}} = 10$ amp.

18) $v = \frac{1}{\sqrt{\mu \in}} = \frac{1}{\sqrt{\mu_0 \varepsilon_0 \mu_r \varepsilon_r}}$

$\Rightarrow v = \frac{C}{\sqrt{\mu_r \varepsilon_r}}$

19) at $t = \frac{T}{12}$,

$\phi = \frac{\pi}{3}$

$x_1 = 0$ and $x_2 = A\sin\frac{\pi}{3} = \frac{A\sqrt{3}}{2}$,

So, $|x_2 - x_1|$

$= \left|\frac{A\sqrt{3}}{2} - 0\right|$

20) $\frac{1}{2}kA^2 = U_{max} - U_{min}$

$\Rightarrow \frac{1}{2}k\frac{400}{10^6} = 0.03$

$\Rightarrow k = \frac{600}{4} = 150\,N/m$

21) $(Slope)_{AD} \propto \gamma$

$\frac{(Slope)_{dia}}{(Slope)_{mono}} = \frac{\gamma_{dia}}{\gamma_{mono}} = \frac{(7/5)}{(5/3)} = \frac{21}{25}$

22) Translational k.E of a molecule $= \frac{3}{2}kT$

Translational k.E of a mole $= \frac{3}{2}RT$

$= \frac{3}{2}PV = \frac{3}{2}\frac{P_0}{8}V_0$

23) Since $t = \frac{\rho L}{2k\theta}\left(x_2^2 - x_1^2\right)$

$t = \frac{\rho L}{2k\theta}\left(x^2 - y^2\right) = \frac{\rho L(x+y)(x-y)}{2K\theta}$

24) $\frac{(V_{rms})_1}{(V_{rms})_2} = \sqrt{\frac{T_1M_2}{T_2M_1}} = 1 \qquad \left[V_{rms} = \sqrt{\frac{3RT}{M_W}}\right]$

$T_1M_2 = T_2M_1 \Rightarrow T_2 = \frac{640 \times 2}{32} = 40\,K$

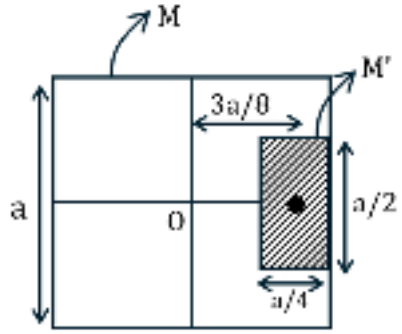
25) Open pipe resonance frequency, $f_1 = \frac{2v}{2L} = \frac{v}{L}$

Closed pipe resonance frequency, $f_2 = \frac{nv}{4L}$,

where n = 1,3,5,.....odd

$f_2 = \frac{n}{4}f_1$ (where n is odd and $f_2 > f_1$), ⇒ n = 5

26) **Solution:**



$$\therefore M' = \frac{M}{a^2} \times \frac{a}{2} \times \frac{a}{4}$$

$$M' = \frac{M}{8}$$

$$X_{com} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

$$X_{com} = \frac{M \times 0 - M' \times r}{M - M'}$$

$$r = \frac{3a}{8}$$

$$X_{com} = \frac{\frac{-M}{8} \times \frac{3a}{8}}{M - \frac{M}{8}}$$

$$X_{com} = \frac{-3a}{56}.$$

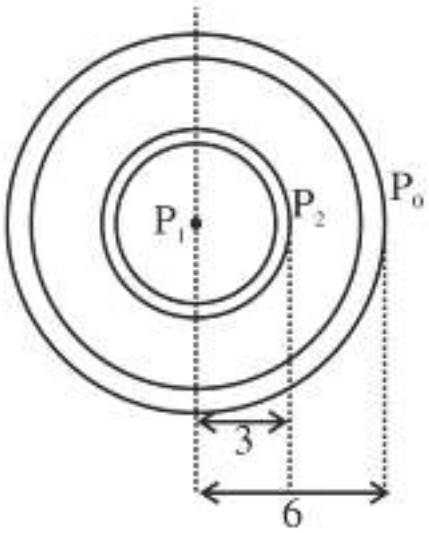
27) wall is parallel to $\hat{j}$.

So, no change in component of velocity along $\hat{j}$.

But $v_x\hat{i} = -eu_x\hat{i}$

$$v_x = -\frac{1}{2}(2) = -1$$

So, $\vec{v} = -\hat{i} + 2\hat{j}$



28)

$$P_2 - P_0 = \frac{4T}{6} \text{ \& } P_1 - P_2 = \frac{4T}{3}$$

$$\Rightarrow P_1 - P_0 = \frac{4T}{2} = \frac{4T}{r}$$

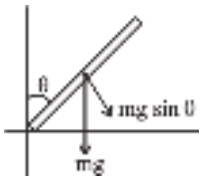
$$\boxed{r = 2}$$

29) $\frac{\Delta U}{V} = \frac{\sigma^2}{2Y}$

here $V = A\ell$ and $\sigma = \frac{F}{A}$

$$\Rightarrow \Delta U = \frac{F^2}{2A^2Y} \times A\ell = \frac{F^2\ell}{2AY}$$

(where, F= 5kg-wt = 50N)

30) 

$\tau = Ia$

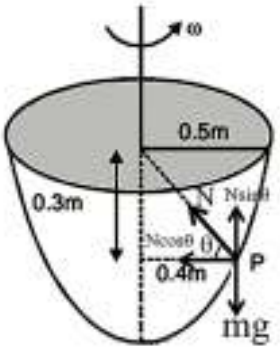
$(mg \sin\theta)\frac{\ell}{2} = \frac{m\ell^2}{3}\alpha$

$\alpha = \frac{3g}{2\ell}\sin\theta$

31) $(T\sin 30° \; L) = (Mg\cos 45°)\frac{L}{2}$

$T \times \frac{1}{2} L = Mg \times \frac{1}{\sqrt{2}} \times \frac{L}{2}$

$T = \frac{Mg}{\sqrt{2}}$

32) 

$\tan\theta = \frac{0.3}{0.4} = \frac{3}{4} \Rightarrow \theta = 37^\circ$

$N \sin\theta = mg$ ..... (1)

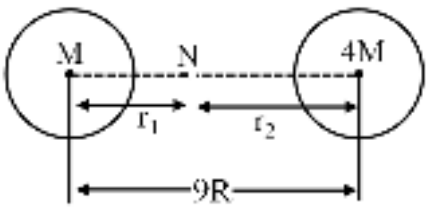
$N \cos\theta = m\omega^2 r$ ....(2)

Equation $(1) \div (2)$

$\tan\theta = \frac{g}{\omega^2 r}$

$\omega^2 = \frac{g}{r\tan\theta} \Rightarrow \omega = \sqrt{\frac{g}{r\tan\theta}}$

$\omega = \sqrt{\frac{10}{(0.4)\frac{3}{4}}} = \sqrt{\frac{100}{3}} = \frac{10}{\sqrt{3}} \text{rad/s}$

33) 

Representation of neutral position (N)

$\therefore \frac{r_1}{r_2} = \sqrt{\frac{m_1}{m_2}}$

$\therefore \frac{r_1}{r_2} = \sqrt{\frac{M}{4M}} = \frac{1}{2}$

$\left.\begin{matrix} r_1 = 3R \\ \Rightarrow r_2 = 6R \end{matrix}\right\}$

By COME law :-

$\frac{1}{2}mv^2 - \frac{GMm}{R} - \frac{G(4M)m}{8R}$

$= -\frac{GMm}{3R} - \frac{G(4M)m}{6R}$

$\frac{1}{2}mv^2 - \frac{12GMm}{8R} = \frac{-6GMm}{6R}$

$\frac{1}{2}mv^2 = \frac{3}{2}\frac{GMm}{R} - \frac{GMm}{R}$

$\frac{1}{2}mv^2 = \frac{GMm}{2R}$

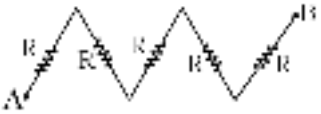
34)

For series combination:

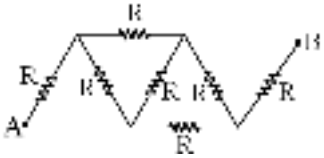
$\frac{1}{C_{eq}} = \frac{d_1}{K_1\varepsilon_0 A} + \frac{d_2}{K_2\varepsilon_0 A} + \frac{d_3}{K_3\varepsilon_0 A} + \frac{d_4}{K_4\varepsilon_0 A}$

$\frac{1}{C_{eq}} = \frac{1}{\varepsilon_0 A}\left(\frac{d_1}{K_1} + \frac{d_2}{K_2} + \frac{d_3}{K_3} + \frac{d_4}{K_4}\right)$

$\frac{1}{C_{eq}} = \frac{\varepsilon_0 A}{\frac{d_1}{K_1} + \frac{d_2}{K_2} + \frac{d_3}{K_3} + \frac{d_4}{K_4}}$

35) 

$R_{AB} = 5R.$



$R_{AB} = 3R.$

Ratio = 5 : 3.

36) $I = \frac{34 - 10}{12} = 2\,\text{amp.}$

$\Rightarrow V_D = 0$

$\therefore V_E = -4 \times 2 = -8$ volt

37)

Before shunting

$\frac{R_1}{R_2} = \frac{40}{60} = \frac{2}{3}$ ..(1)

After shunting

$\frac{R_1}{R'_2} = \frac{50}{50} = \frac{1}{1}$ ..(2)

from eq. (1) & (2)

$\frac{R'_2}{R_2} = \frac{2}{3}$

Now $R'_2 = \frac{R_2(10)}{R_2 + 10}$

so $\frac{10}{10 + R_2} = \frac{2}{3} \Rightarrow$ $R_2 = 5\Omega$

From eq. (i)

$R_1 = R_2\left(\frac{2}{3}\right) = \frac{10}{3}\ \Omega$

38)

$[\beta] = [FV][x^2]$

$= [M^1L^1T^{-2}]\ [LT^{-1}][L^2] = [M^1L^4T^{-3}]$

39) $L.C. = \frac{1}{50}mm$ = 0.02 mm

Zero error = 3 × 0.02 = 0.06 mm

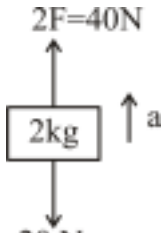
Reading = 7 mm + (36 × 0.02 mm)

= 7.72 mm

Correct value = M.V. – Z.E.

= 7.72 – 0.06

= 7.66 mm



40) $a = \frac{40 - 20}{2}$

$a = 10\ m/sec^2$

41) Total mass of 80 wagons = $80 \times 5 \times 10^3 = 4 \times 10^5$ kg

$a = \frac{F}{M} = \frac{4 \times 10^5}{4 \times 10^5} = 1 m/sec^2$

Tension in the coupling between 30th and 31th wagons will be due to mass of remaining 50 wagons. Now, mass of remaining 50 wagons

$m = 50 \times 5 \times 10^3 = 25 \times 10^4$ kg

Required tension T = ma
$= 25 \times 10^4 \times 1 = 25 \times 10^4$ N

42) F = 0.2 x retardation, i.e.

$$mv\frac{dv}{dx} = -0.2x$$

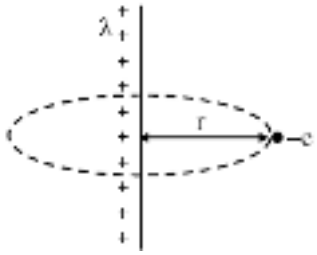
$$\Rightarrow m\int_{u=10}^{v} vdv = -0.2\int_{x=0}^{x=10} xdx$$

$$\frac{1}{2}m(v^2 - u^2) = -0.2\left(\frac{x^2}{2}\right)$$

$$\frac{1}{2}mv^2 - \frac{1}{2} \times 10 \times 10^2 = -\frac{1}{10}(10^2 - 0) = -10$$

$$\frac{1}{2}mv^2 = 500 - 10 = 490\,J$$

43)



Electric field E at a distance r due to infinite long wire is $E = \frac{2k\lambda}{r}$

Force of electron ⇒ F = eE

$$F = e\left(\frac{2k\lambda}{r}\right)$$

$$F = \frac{2k\lambda e}{r}$$

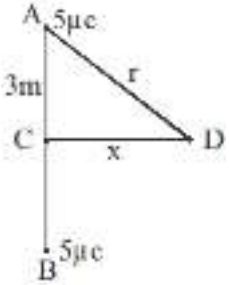
This force will provide required centripetal force

$$F = \frac{mv^2}{r} = \frac{2k\lambda e}{r}$$

$$v = \sqrt{\frac{2k\lambda e}{m}}$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{2k\lambda e}{m}\right) = k\lambda e$$

This is constant so option (B) is correct



44)

KC + UC = KD + UD

$$0.06 + \left(\frac{K \times 5 \times 10^{-6}}{3} \times 2\right) \times \left(-5 \times 10^{-6}\right)$$

$$= 0 + (-5 \times 10^{-6}) \times \frac{K \times 5 \times 10^{-6}}{r} \times 2$$

$\Rightarrow$ r = 5m

$\Rightarrow$ $x = \sqrt{r^2 - 3^3} = 4m$

45)

Statement 1 suggests mutual inductance can only be found if current in the small coil varies, which is incorrect
Mutual inductance is a property of the pair of coils regardless of which one carries the varying current
Statement 2 correctly describe the reciprocity theorem $M_{12} = M_{21}$ and its dependence on geometric factors
Only Statement 2 is correct.

CHEMISTRY

46)

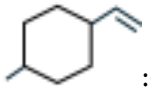
CH_3–CHD–CH_2T, CH_2D–CHT–CH_3, CHDT–CH_2–CH_3
1 2 3
1 & 2 → Positional isomers
2 & 3 → Positional isomers

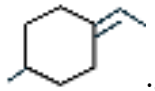
47)

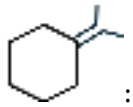
Stereo isomers = 2^n (n ⇒ G. I. site or chiral site)
= 2^3 = 8

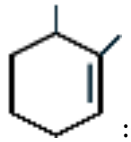
48) **Solution:**
A. Chloroform & aniline → Distillation (P) Both are liquids with different boiling points → separated by **Distillation**.
B. Aniline & water → Steam distillation (R) Aniline is steam-volatile and immiscible with water → separated by **Steam Distillation**.
C. Amino acid → Chromatography (Q) Amino acids are polar solids, best separated/purified by **Chromatography**.
D. Glycerol → Distillation under reduced pressure (S) Glycerol has very high boiling point and decomposes on heating → **purified by Vacuum Distillation**.
Correct matching is A-P; B-R; C-Q; D-S
Final Answer: Option (3).

49) **Concept:** Heat of hydrogenation $\propto \frac{1}{\text{stability of alkene}}$
Solution:

[P]  :- ∝ – C – H ⇒ 1

[Q]  :- ∝ – C – H ⇒ 7

[R]  :- ∝ – C – H ⇒ 10

[S]  :- ∝ – C – H ⇒ 6

 stability of Alkene is :- R > Q > S > P

 order HOH is :- P > S > Q > R.

Final Answer:

Option (1)

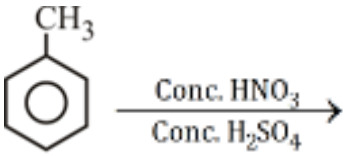
50)

Anti-Aromatic decide by rules

51)

Acid which is more acidic then H_2CO_3 react with $NaHCO_3$

52)

(a)  (q) Nitration

(b) CH_3–COOK $\xrightarrow{\text{electrolysis}}$ (p) Kolbe's electrolysis

(c) CH_3–Br $\xrightarrow[\text{Dryether}]{\text{Na}}$ (r) Wurtz reaction

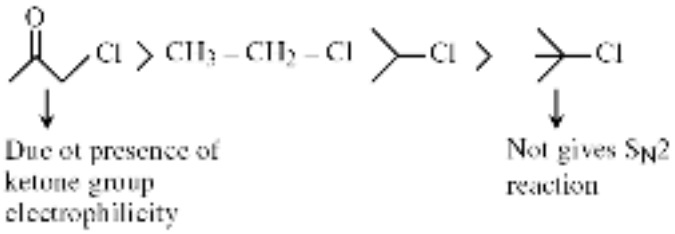
(d)  $\xrightarrow[\text{Anhyd. } AlCl_3]{CH_3Cl}$ (s) Friedal-craft alkylation

53)

Solution :-

Rate of S_N2 reaction ∝ electrophilicity of R – X of $\propto \frac{1}{\text{Steric crowding}}$

– I and –M group on alkyl halide increases electophilicity



Final Answer :- (4)

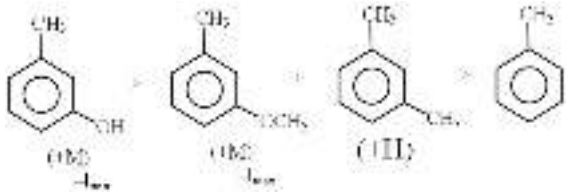
54)

$SOCl_2$ → SN^i mech.
PCl_3 → by S_N2 mech.

55)

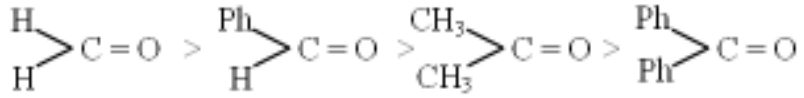
Solution :-

(a)



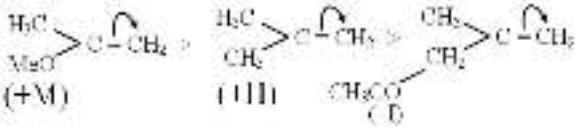
Rate of ESR ∝ e^- density on ring < $\frac{EDG}{EWG}$
Given order is correct.

(b)



Rate of NAR ∝$^{\oplus}$ charge on carbonyl carbon ∝ $\frac{1}{\text{Steric hinderance}}$
Given order is correct.

(c)



Rate of EAR ∝ stability of $C^{\oplus}$
Given order is correct.
(d) Rate in F.S.R. α stability of free radical
So all a,b,c,d are correct
Final Answer :- (4)

56)

NCERT name reaction

57)

Rate in S_NAE ∝ Leaving ability
∝ more δ⊕ on carbonyl carbon (if leaving group is same)

58)

$CH_3–CH_2–N\equiv C \xrightarrow{H_2/Pd–C} CH_3–CH_2–NH–CH_3$

59) Theory based

60)

More – M, – H, – I → More acidic

61) $N_{(g)} + e^- \rightarrow N_{(g)}^-$ (Process is endothermic)
Hence, it's opposite process is exothermic
$N^-_{(g)} \rightarrow N_{(g)} + e^-$

62)

Option	Order given	Correct idea	Status
1	$Cl_2O_7 > SO_3 > P_4O_{10}$	Non-metal oxides, acidity order correct	✓
2	$N > O > P > S(IE)_1$	First ionization energy trend valid	✓
3	$Mn_2O_7 > Mn_2O_3 > MnO_2$	$Mn_2O_7 > MnO_2 > Mn_2O_3$	✗
4	$Li^+(aq) > Na^+(aq) > K^+(aq)$	Hydrated ionic radius order correct	✓

63)

Correct order of electron affinity is

S > Se > Te > O

64) Adding a second electron to an anion is endothermic due to electron–electron repulsion. Smaller O^- has greater repulsion than larger S^-, so O shows higher positive value than S.

65)

HOCl is most Powerful oxidising agent
Basic nature ⇒ $NH_3 > PH_3 > AsH_3$
BP ⇒ $NH_3 > AsH_3 > PH_3$
Acidic nature ⇒ HF < HCl < HBr < HI

66) $[M(AA)_2a_2]$ → show GI and OI both
$[M(AA)a_4]$ → do not show GI and OI both
$[M(abcd)] \xrightarrow{\text{tetrahedral}}$ do not show GI, but show OI
$[Mabcd] \xrightarrow{\text{Square planar}}$ show GI but do not show OI

67)

Turnbull blue = $Fe_3\,[Fe(CN)_6]_2$

68) $E = \frac{hc}{\lambda}$

69)

$K_4[\,Fe(NO_2)_2(CN)_4\,]$
$Fe^{+2} = 3d^6 \quad t^6_{2g}\, e^0_g$
$[\,Ni(NH_3)_2\,(en)_2]SO_4$
$Ni^{+2} = 3d^8$ $\boxed{n = 2}$, octahedral

70)

Only VF_5 and VCl_5 exist

71)

Mn^{+2} has max unpaired e^-

72)

Melting point depends on metallic bonding and number of unpaired d-electrons.
• Cr has half-filled d-orbitals → verry
strong melting bonding → high melting point.
• V ($3d^3 4s^1$) has fewer unpaired electrons →
lower melting point than Cr.
• Mn ($3d^5\ 4s^2$)has weaker metallic bonding →
lower melting point than Cr.
• Ti ($3d^2\ 4s^2$) has lower melting point than V.

73)

V^{+4}, Cu^{+2} = blue
Ni^{+2} ⇒ Green
Fe^{+3} ⇒ yellow
Cr^{+3} ⇒ violet

74)

$H_2S_2O_6$, $H_2S_2O_4$, $H_2S_2O_5$ have S–S bond.

75) For Fe^{3+}, $A\square^{3+}$, Cr^{3+}, with NH_4OH excess NH_4Cl is added.

76)

Theory based

77)

(1) = (2) + (3) rev
–a = (–b) + (+c)
a = b–c

78)

$Kt = 2.303 \log \frac{p_0}{p_t}$
$t_{1/2} = \frac{0.693}{K}$

79)

$\Sigma G_P - \Sigma G_R = \Delta G° = -RT \ln K$
$\left(2 \times G_{NO_2}\right) - (2 \times 86600) = - R \times 298 \ln 1.6 \times 10^{12}$
$G_{NO_2} = \frac{1}{2}(2 \times 86600 - R \times 298 \ln 1.6 \times 10^{12})$
$= 0.5 (2 \times 86600 - R \times 298 \ln 1.6 \times 10^{12})$

80)

Total moles = 1 + 0.5 = 1.5
Total mass of solvent = 51 + 2 = 53
$\text{molality} = \frac{1.5}{53} \times 1000 = 28.3$

81)

$Cu^{2+} + 2e^- \rightarrow Cu$
2F → 1 mol
1F – 0.5 mol
mole of Cu^{2+} left in sol. = 1.5 – 0.5 = 1 mol
$Ag^+ + e^- \rightarrow Ag$
1F → 1 mol
0.1 F → 0.1 mol
mole of Ag^+ left in sol = 0.2 – 0.1 = 0.1
cell reaction : $Cu + 2Ag^+ \rightarrow Cu^{2+} + 2Ag$
$E^o_{cell} = SRP)_C - SRP)_A$
= 0.8 – 0.34
= 0.46
$E_{cell} = 0.46 - \frac{0.06}{2} \log \left(\frac{1}{(0.1)^2}\right)$

$= 0.46 - \frac{0.06}{2} \times 2$

$= 0.46 - 0.06$

$= 0.4$

82)

Theory based

83) $\Delta T_b = iK_bm$

$2 = \left(1 - \frac{1}{2} \times 0.8\right) \times 5 \times \frac{W}{122 \times 30} \times 1000$

W = 2.4

84) x mol SO_2Cl_2

$\square\ n_{H_2SO_4} = x$

$n_{HCl} = 2x$

for H_2SO_4

x mol H_2SO_4 required 2x mol of NaOH for HCl

2x mol HCl required 2x mol of NaOH

Total mol of NaOH = 4x = 16

x = 4

85) $K = \frac{2^2}{2 \times 2} = 1$

$A_2 + B_2 \rightleftharpoons 2AB$

t = t 3 3 2

$t = t_{eq}$ 3–x 3–x 2+2x

$K = \frac{(2+2x)^2}{(3-x)^2} \Rightarrow \frac{2+2x}{3-x} = 1 \Rightarrow 2 + 2x = 3 - x$

3x = 1

$x = \frac{1}{3}$

86) $2 \times E_{Cu/Cu^{2+}} = 1 \times E^o_{Cu/Cu^+} + 1 \times E^o_{Cu^+/Cu^{2+}}$

$E_{Cu/Cu^{2+}} = \frac{-0.35 - 0.25}{2}$

= – 0.3

$E^o_{Cell} = SRP)_C - SRP)_A$

= (0.8) – (0.3) = 0.5 V

n = $2e^-$

$E^o_{Cell} = \frac{0.0591}{2} \log K \Rightarrow K = 8 \times 10^{16}$

87) $\alpha = \frac{\lambda_M}{\lambda^o_M}$

$$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{C\left(\frac{\lambda_M}{\lambda_M^0}\right)^2}{1-\frac{\lambda_M}{\lambda_M^0}}$$

88)

$FeSO_4$, FeC_2O_4 , $Fe(NO_3)_2$, $KMnO_4$

V.F = 1 V.F = 3 V.F = 1 V.F = 5

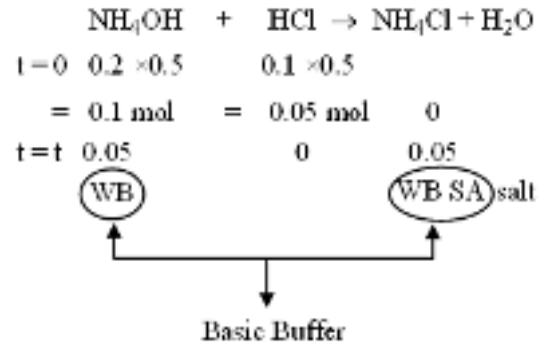
$x \times 5 = 1 \times 1$ | $y \times 5 = 1 \times 3$ | $z \times 5 = 1 \times 1$

$x = \frac{1}{5}$ | $x = \frac{3}{5}$ | $x = \frac{1}{5}$

total mol of $KMnO_4$ = x + y + z

$= \frac{1}{5} + \frac{3}{5} + \frac{1}{5} = 1$

89)



$$pOH = pK_b + \log\left(\frac{0.05}{0.05}\right) = pK_b = 4.75$$

⇒ pH = 9.25

90)

Theory based

BIOLOGY

91)

NCERT XII Pg. No. # 198, 199, (E) 217, 218 (H)

92) *Explanation:* Parasites do not always kill their hosts (Statement B is incorrect), and parasitism never benefits the host (Statement D is incorrect).

93)

NCERT XII Pg. No. # 205, 207, 209, 214 (E)
224, 226, 228 (H)

94) NCERT XII Pg. # 207, 209 (E), 226, 228, 229 (H)

95) NCERT XII Pg. No. # 212 (E), 232 (H)

96) NCERT XII Pg No.- 222 (E), 242, 243 (H)

97) NCERT XII Pg No. - 225 (E), 245 (H)

98) NCERT XII Pg No. 221, 222, 223, 225 (E)
242, 243, 245 (H)

99) **Explain Question :**The question asks how many of the given statements about RNA are correct.

Concept : RNA, Types Of RNA

Explanation :
Statement I: True — single RNA polymerase transcribes all RNA types in bacteria.

Statement II: True — the three major RNA types are correct.

Final answer : Option (1)

100) **Explain Question :**The question asks about the function of mRNA in cells.

Concept : RNA, Types Of RNA

Explanation : The primary function of mRNA is to provide the template for protein synthesis.

Final answer : Option (2)

101)

Explain Question : The question asks which statement about DNA replication experiments is not correct.

Concept : DNA replication experiments

Explanation : Meselson and stall proposed a scheme for replication of DNA. - Incorrect – Meselson & Stahl did not propose a replication scheme; they proved the semiconservative model.

Final answer : Option (1)

102) NCERT Pg.# 103

103)

Explain Question: The question is asking you to check if both the Assertion (A) and Reason (R) are true or false, and whether the Reason correctly explains the Assertion about the steps of translation, especially amino acid activation and peptide bond formation.

Concept: Translation.

Solution:

The Correct Answer: (3) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion

Explanation :

Statement	**Truth**	**Explanation**
Assertion: Amino acids are activated in the presence of ATP during the first phase of translation.	True — Amino acids are activated (aminoacyl-AMP formed) using ATP before attaching to tRNA.	
Reason: When two charged tRNAs are close, peptide bond formation is energetically favored.	True — Ribosome catalyzes peptide bond formation between amino acids carried by charged tRNAs.	
Relation:	Not a correct explanation — Reason describes peptide bond formation (elongation phase), while Assertion describes amino acid activation (initiation stage).	

Hence, both are correct, but Reason does not explain Assertion → Option (3).

Final answer: Option (3) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion

104)

NCERT XII Pg No # 85, 86, 87

105) NCERT-Pg. # 69,70

106)

NCERT XII Pg. No. # 61, 62, 63

107)

NCERT XI Pg. No. # 61

108) NCERT, Pg # 70,71

109) NCERT XI, Pg. # 175

110) NCERT XI Pg. # 174

111) NCERT XI g. # 179

112) NCERT XI Pg. # 159,160,161

113) NCERT-XI, Pg. No. # 164, 165

114)

NCERT XI Pg. No. # 135

115) NCERT XI Pg. # 145, 146

116) NCERT XI Pg. # 108

117) NCERT XI, Pg # 105

118) NCERT XI Pg. # 118

119) NCERT XI Pg # 116, 117

120) NCERT. Pg. No. 74

121) NCERT Page No. 58

122) NCERT page No. - 63,64,65

123) NCERT Page No. 64

124) NCERT Pg.#74,77.

125) NCERT-XII, Pg # 6

126) NCERT-XII_Pg.# 19

127) NCERT-XII_ Pg.#11,12.

128) NCERT Pg.#5, 6

129) NCERT Pg. # 14

130) NCERT Pg. # 20, 21

131) NCERT-XI, Pg. # 14, 15, 16, 18

132) NCERT Pg#9

133) NCERT Pg. # 27

134)

NCERT XI - P. No. - 39,41

135) NCERT Pg. No. # 33 & 34

136) NCERT Pg.#42-45.

137)

NCERT Pg. # 42, 44

138) NCERT Pg. # 57 (Para # 3), Module-4 Pg. # 51

139) NCERT, Pg. # 44

140) NCERT, Pg. # 46

141) NCERT Pg. # 114

142)

NCERT (old) Pg. # 104

143) NCERT Pg. # 83

144)

NCERT Pg. # 242, 243

145) NCERT Pg # 242, 243

146) NCERT Pg. # 241

147) NCERT Pg. # 236

148) NCERT Pg. # 317

149) NCERT Pg. # 227

150) NCERT Pg. 222

151) NCERT Pg. # 199, 200

152) NCERT Pg. # 195

153)

NCERT Pg. # 199

154)

NCERT Pg. # 209

155) NCERT Pg. No. # 213

156) NCERT Pg. # 186

157) NCERT Pg. # 190

158) NCERT, Pg. # 35

159) NCERT Pg. # 32

160) NCERT Pg. # 54

161) NCERT Pg. # 44

162) NCERT Pg. No. # 148

163) NCERT Pg. # 144

164) NCERT Pg # 137, 138

165) NCERT Pg. # 137

166) NCERT Pg. # 142

167) NCERT Pg. 188, 119

168) NCERT Pg.# 120, 121

169) NCERT Pg. # 115

170) NCERT Pg.# 171

171) NCERT-XII, Pg # 195, 198, 202, 208

172) NCERT XII Pg. # 164

173)

NCERT XII Pg. # 182

174) MODULE Pg. No. # 12

175) NCERT Pg. # 183, 184

176) NCERT XI Pg # 134, 8.5.3.2

177) NCERT, XI Pg # 98

178) NCERT, XI Pg # 88

179) NCERT, XI Pg # 125 to 127

180) NCERT, XI Pg # 122 to 124